TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering

The Zandman-Slaner School of Graduate Studies

Low-Power Arithmetic for Configurable Multi-Rate Digital Interpolation Filters, from MATLAB to GDSII: Polyphase, Cubic and Spline Compared on Two TSMC 65 nm Tapeouts

A thesis submitted towards the degree of
Master of Science in Electrical Engineering

by

Mohammad Khamaysi

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Abstract

A digital transmitter that hands QAM-64 samples to a converter at 60 MSPS reproduces its spectrum every 60 MHz, and the analogue filter that must remove the first image is steep, expensive and temperature-sensitive. Interpolating digitally before the converter moves that image away and trades the analogue problem for digital arithmetic — specifically for the multipliers of the interpolation filter. This work is about that arithmetic: which interpolation architecture gives the best EVM and replica level for a given multiplier count, what it costs to remove the multipliers altogether by hardwiring the coefficients as shift-add trees, and how a design is carried from a mathematical model to signed-off silicon without a single number in the record that cannot be traced to the tool report or measurement it came from.

Three interpolation architectures — a polyphase FIR, a cubic interpolator and a recursive MinAJ2 Hermite spline — share each of two dies in TSMC 65 nm LP, behind one runtime-reconfigurable interface covering four interpolation factors and three word lengths at up to 960 MHz. The mathematical development covers the polyphase decomposition, the per-factor band-edge design of the equiripple prototype, the cubic and Hermite kernels with their fixed-point weight tables, the accumulator bounds derived per core, and the signed-power-of-two substitution; every equation was checked against the MATLAB model and the RTL that built the chips. One chip keeps every coefficient loadable; the other hardwires every coefficient, and is smaller in every synthesis count.

The design flow is documented stage by stage — MATLAB golden model, RTL, synthesis, pad ring, floorplan, placement, clock tree, routing, post-layout timing, post-layout SDF simulation, DRC, LVS, equivalence checking, IR drop, timing sign-off and GDS streamout — with the report of each stage reproduced from the tool's own output and the layout captured from the final databases. One golden model is compared bit for bit at every representation of the design, and both chips are bit-exact in post-layout simulation at every legal mode and both characterisation corners; both were submitted for fabrication in September 2026.

Every number in this thesis is traced to the report, log, simulation compare or instrument readback it was read from; none is estimated, and no measured-silicon result is claimed.

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Abbreviations

ASIC	application-specific integrated circuit
CFG / NCFG	the loadable-coefficient chip / the hardwired-coefficient chip
CSD	canonical signed digit
CTS	clock-tree synthesis
DAC	digital-to-analogue converter
DC	Design Compiler (Synopsys), the synthesis tool
DRC / LVS / ERC	design-rule check / layout-versus-schematic / electrical-rule check
EVM	error vector magnitude
FIR	finite impulse response
FSDB	fast signal database (Verdi waveform file)
GDS	the mask database streamed to the foundry (GDSII)
GLS	gate-level simulation
ICG	integrated clock-gating cell
IR drop	supply voltage loss along the power grid
LEC	logic equivalence check (Cadence Conformal)
MER	modulation error ratio
MPW	multi-project wafer
P&R	place and route (Cadence Innovus)
PT	PrimeTime (Synopsys), the timing sign-off tool
QoR	quality of results (synthesis report)
RRC	root-raised cosine
RTL	register-transfer level (SystemVerilog)
SAIF	switching-activity interchange format
SDC	Synopsys design constraints
SDF	standard delay format
SPT	signed power of two
STA	static timing analysis
VSA	vector signal analyser (Keysight 89600)
WL / L	input word length / interpolation factor

Chapter 1

Background and motivation

A transmitter hands its samples to a converter at f_s, and the converter reproduces the spectrum around every multiple of f_s,

$$X_s(f) = \sum_{m=-\infty}^{\infty} X(f - m f_s). \tag{1}$$

For QAM-64 at 20 MBd with roll-off 0.15, sampled at $f_s = 60$ MHz, the wanted band ends at 11.5 MHz and the first image begins at 48.5 MHz, so the analogue reconstruction filter must be steep. Interpolating digitally by L removes the images below $L f_s$ and makes the analogue filter simple (Fig. 1). The cost moves to the interpolation filter's multipliers, and that arithmetic is the subject of this work.

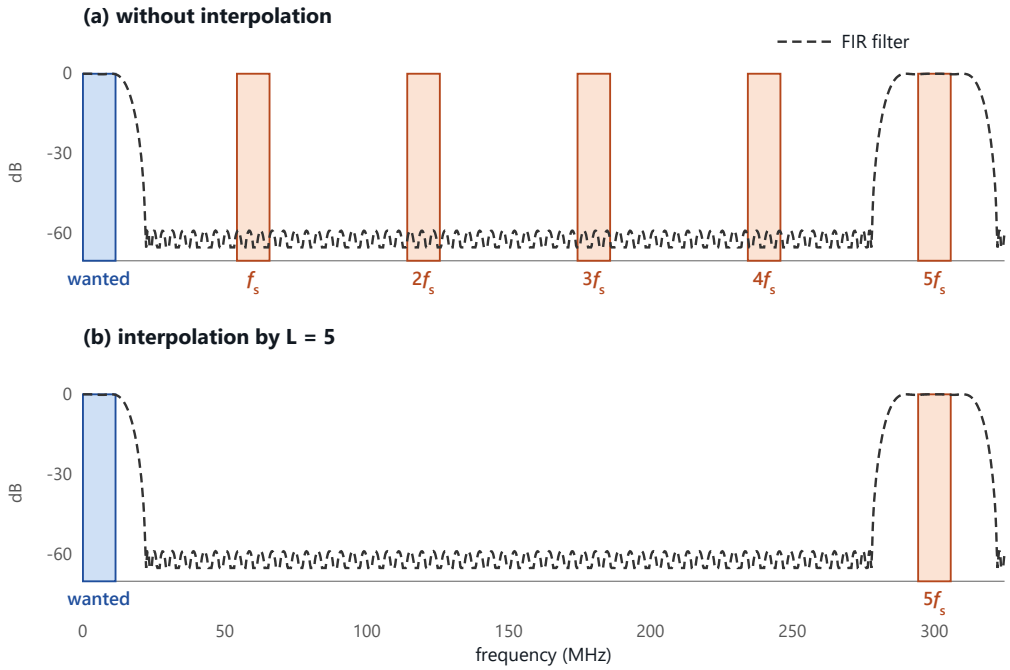


Figure 1. Why interpolate. The dashed line is the interpolation filter, an FIR low-pass that passes the wanted band and is the same in both panels. (a) Without interpolation the images sit at every multiple of f_s, the first one next to the wanted band, and the analogue filter has to separate them. (b) Interpolating by $L = 5$ removes the images at f_s, $2f_s$, $3f_s$ and $4f_s$; the first remaining image is at $5f_s$, far from the wanted band.

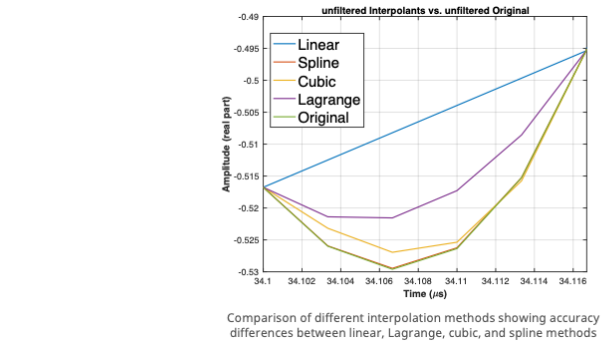
1.1 Three architectures and one substitution

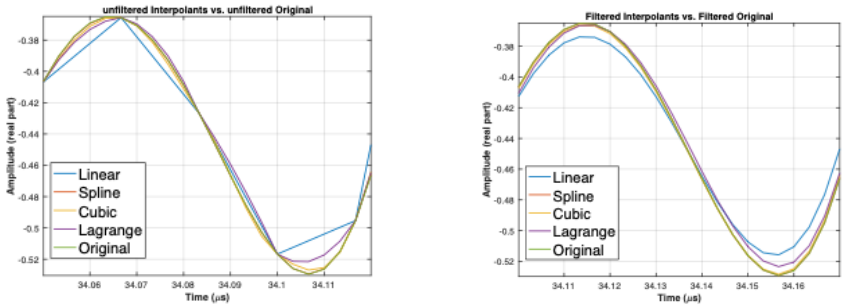
The *polyphase FIR* splits a filter designed at the output rate into L sub-filters at the input rate; it is minimax-optimal for its length and its quality is set by the prototype's band edges. *Cubic* interpolation fits a polynomial through four neighbouring samples at four multiplications per output and needs a short cleanup filter. The *MinAJ2 Hermite spline* picks the slope at each knot to minimise the integrated

squared third derivative, a one-sided recursion evaluated in Horner form. All three share each die. A constant multiplication can be replaced by shifts and additions,

$$c = \sum_{i=1}^{T} s_i \, 2^{e_i}, \qquad s_i \in \{+1, -1\}, \tag{2}$$

so if every coefficient is a constant, every multiplier disappears; what that buys and what it costs is measured here on silicon. Fig. 2 shows the interpolators side by side in the MATLAB model, before and after the cleanup filter.





(a) unfiltered interpolants and the original **(b)** after the cleanup filter

Figure 2. The interpolators compared in the MATLAB model, from the earlier study that chose the cores. Top: one interval between two input samples; the spline and the cubic track the original, the linear interpolant cuts the corner. Bottom: a full cycle before and after the cleanup filter.

1.2 Figures of merit and contributions

EVM is how far each received symbol lands from its constellation point; replica level is how far the strongest surviving image sits below the signal. They trade against each other through the filter order, and both are read from a Keysight 89600 vector signal analyser, never computed. The contributions are: a same-die comparison of three architectures against one golden model, with a per-factor band-edge design that improved both figures of merit at no hardware cost; two tapeouts in TSMC 65 nm LP, loadable and hardwired coefficients, documented from the tool reports; and one verification method, bit-exact at every representation. No silicon has returned yet, and no measured-silicon result is claimed.

Chapter 2

Signal model and mathematical development

2.1 The signal and the operating modes

Symbols at $R_s = 20\,\text{MHz}$, roll-off $\alpha = 0.15$, three samples per symbol at $f_s = 60\,\text{MHz}$:

$$B = \frac{(1+\alpha)\,R_s}{2} = 11.5\,\text{MHz}, \qquad f_s - B = 48.5\,\text{MHz}. \tag{3}$$

Every filter must pass 11.5 MHz and stop by 48.5 MHz. The serial clock is WL $\times$ 60 MHz and the output rate $L \times 60$ MHz; seven combinations are legal (Table 1). The datapath is 16-bit at every word length; shorter words are shifted up so that the sign lands on bit 15. One QAM-64 stream, generated once with a recorded seed and cut into one segment per legal mode plus two illegal headers, is the stimulus at every stage, and its expected output comes from the same model.

Table 1. Legal operating modes. DESIGN-PARAM (the `modes` table in `final_script.m`).

WL (bits)	L	serial clock (MHz)	output rate (MHz)	input alignment	segment
12	2	720	120	$\ll 4$	seg01
12	3	720	180	$\ll 4$	seg02
12	4	720	240	$\ll 4$	seg03
15	3	900	180	$\ll 1$	seg04
15	5	900	300	$\ll 1$	seg05
16	2	960	120	native	seg06
16	4	960	240	native	seg07

2.2 Polyphase FIR interpolation

Interpolation by L is zero-stuffing followed by a low-pass filter at the output rate. Let $v[m]$ be the zero-stuffed signal: $v[m] = x[m/L]$ when L divides m and $v[m] = 0$ otherwise. Filtering it with the prototype h of $N = LP$ taps is the convolution

$$y[m] = \sum_{k=0}^{N-1} h[k]\, v[m-k]. \tag{4}$$

Write the output time as $m = nL + p$, with n the input-rate time and $p \in \{0, \ldots, L-1\}$ the phase, and the tap index as $k = rL + q$ with $q \in \{0, \ldots, L-1\}$:

$$y[nL+p] = \sum_{r=0}^{P-1} \sum_{q=0}^{L-1} h[rL+q]\; v\big[(n-r)L + (p-q)\big]. \tag{5}$$

The index of v is a multiple of L only when $q = p$. Every other term multiplies a stuffed zero and vanishes, so the double sum collapses to a single one over the input samples,

$$y[nL+p] = \sum_{r=0}^{P-1} h_p[r]\, x[n-r], \qquad h_p[r] = h[rL+p], \tag{6}$$

L sub-filters of $P = 7$ taps running at the input rate and a commutator that emits phase p at output time $nL + p$ (Fig. 3). No multiplication by a zero is ever performed: the LP-tap filter costs LP multiplications per input sample instead of the L^2P of direct convolution at the output rate. The prototype is scaled by L so that each phase has unit DC gain and one right shift of 15 is correct at every L. The core computes all five phases in parallel, 35 multipliers per rail in Chip 1 (CFG); in Chip 2 (NCFG) the same products are shift-add trees.

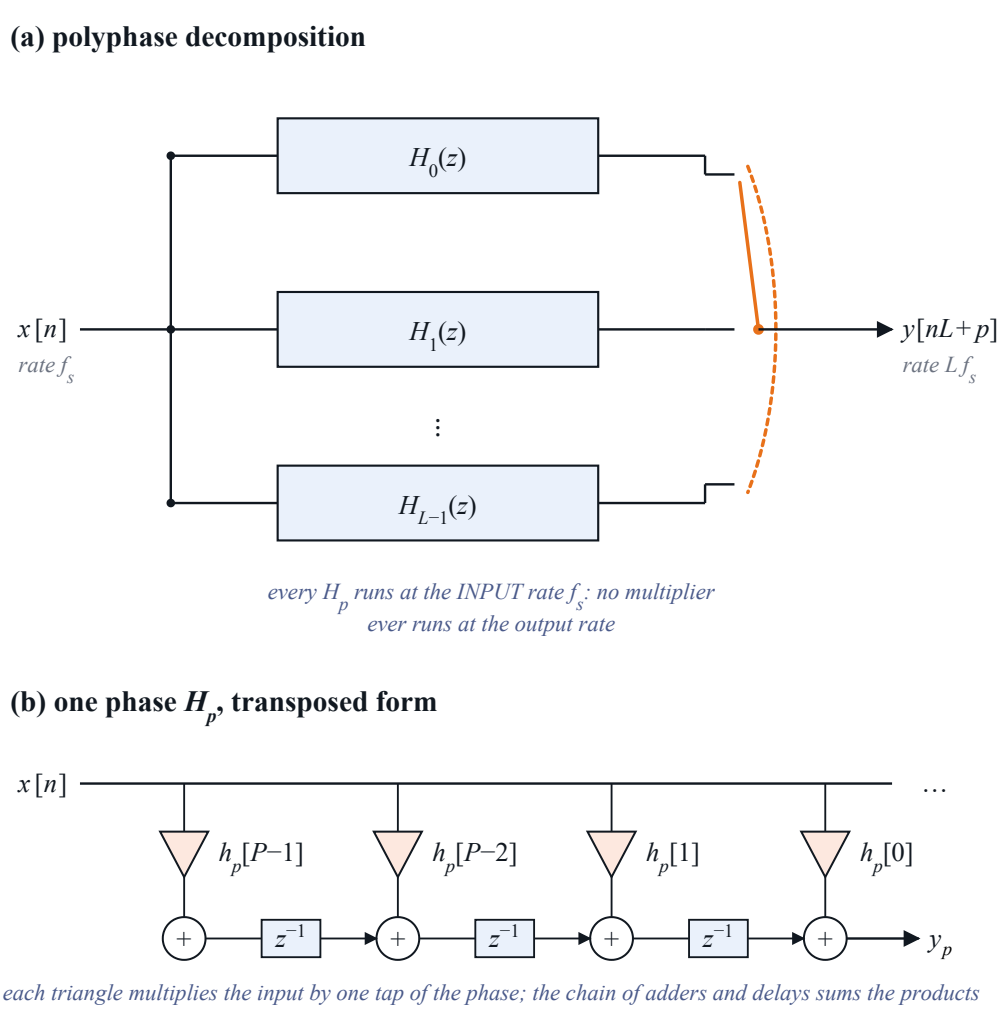


Figure 3. Polyphase decomposition. (a) Each of the L phases H_p is a short FIR at the input rate; the commutator selects phases in turn. (b) One phase in transposed form: the input is broadcast to all P coefficients and the products are summed through a delay chain.

2.3 MinAJ2 recursive Hermite spline

A classical cubic spline passes one piecewise cubic through all the samples with continuous first and second derivatives, but it needs a global linear system and is not causal. The chip uses a local method. Between two samples x_i and x_{i+1} the segment is a cubic Hermite polynomial in the fractional delay $u = (t - t_i)/h \in [0, 1]$,

$$S_i(u) = H_{00}(u)\, x_i + H_{01}(u)\, x_{i+1} + H_{10}(u)\, m_i + H_{11}(u)\, m_{i+1}, \tag{7}$$

with the Hermite basis

$$H_{00} = 2u^3 - 3u^2 + 1, \qquad H_{01} = -2u^3 + 3u^2, \qquad H_{10} = u^3 - 2u^2 + u, \qquad H_{11} = u^3 - u^2, \tag{8}$$

and $m_i = h\, s_i$ the slope at knot i scaled by the sample spacing h. Every segment is a cubic, so the interpolant is continuous with a continuous first derivative everywhere; the second derivative is continuous across a knot only if the slopes are chosen well.

MinAJ2 chooses m_i at each knot to minimise the accumulated squared jerk, the third derivative, over the two segments that meet at the knot,

$$J(m_i) = \int_{t_{i-1}}^{t_i} \big(S'''_{i-1}\big)^2\, dt + \int_{t_i}^{t_{i+1}} \big(S'''_i\big)^2\, dt. \tag{9}$$

The third derivative of a cubic is a constant on its segment,

$$S'''_{i-1} = \frac{6}{h^3}\big[\, 2(x_{i-1} - x_i) + m_{i-1} + m_i \,\big], \qquad S'''_i = \frac{6}{h^3}\big[\, 2(x_i - x_{i+1}) + m_i + m_{i+1} \,\big]. \tag{10}$$

The right segment still holds the unknown m_{i+1}. It is removed by asking for a continuous second derivative at the knot, $S''_{i-1}(t_i) = S''_i(t_i)$, which gives $m_{i+1} = 3x_{i+1} - 3x_{i-1} - 4m_i - m_{i-1}$ and therefore

$$S'''_i = \frac{6}{h^3}\big[\, -3x_{i-1} + 2x_i + x_{i+1} - 3m_i - m_{i-1} \,\big]. \tag{11}$$

With (10) and (11) in (9), J is a quadratic in m_i; setting $dJ/dm_i = 0$ gives the slope recursion

$$m_i = \frac{-11\, x_{i-1} + 8\, x_i + 3\, x_{i+1} - 4\, m_{i-1}}{10}, \tag{12}$$

which needs only the three samples around the knot and the previous slope, so it streams. For evaluation the segment is written in the power basis and evaluated in Horner form,

$$S_i(u) = \big((c_3\, u + c_2)\, u + c_1\big)\, u + c_0, \tag{13}$$

$$\begin{aligned} &c_0 = x_i, && c_1 = m_i, && \Delta = x_{i+1} - x_i, \\ &c_2 = 3\Delta - 2m_i - m_{i+1}, && c_3 = m_i + m_{i+1} - 2\Delta, \end{aligned} \tag{14}$$

three multiply-add stages (Fig. 4). In the chip the division by ten is one constant multiplication, and every product is rounded back to 16 bits.

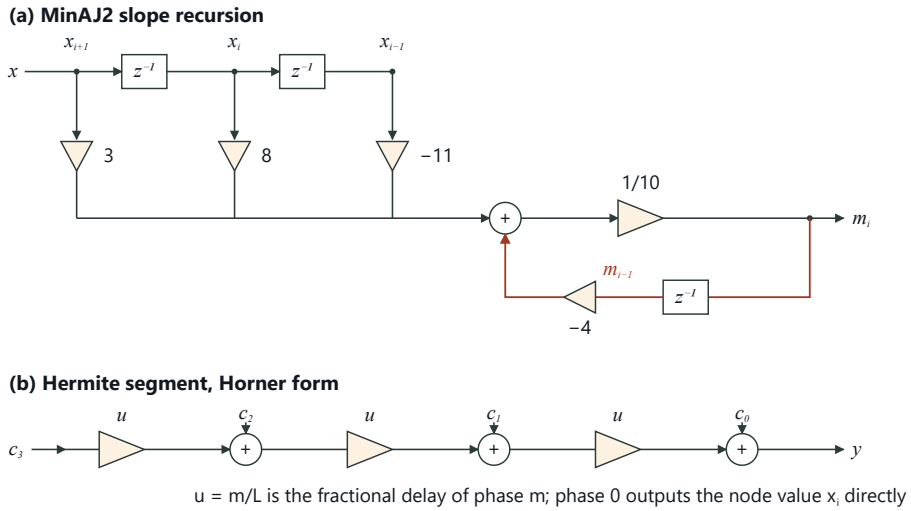


Figure 4. MinAJ2 Hermite spline core. (a) The slope recursion of (12): three weighted samples and the previous slope fed back through a delay. (b) The Hermite segment in Horner form, three identical multiply-add stages.

2.4 Cubic interpolation

A cubic through four samples at $-1, 0, 1, 2$, evaluated at $u = m/L$, has the weights

$$\mathbf{w}(u) = \left[-\frac{u^3}{6} + \frac{u^2}{2} - \frac{u}{3}, \quad \frac{u^3}{2} - u^2 - \frac{u}{2} + 1, \quad -\frac{u^3}{2} + \frac{u^2}{2} + u, \quad \frac{u^3}{6} - \frac{u}{6} \right]. \tag{15}$$

The weights are constants once L is fixed and are held as a Q15 table, each row rounded by largest remainder to sum to exactly 32768 so that the phases cannot amplitude-modulate one another: at $u = \frac{1}{2}$ the row is $[-2048,\ 18432,\ 18432,\ -2048]$. Phase 0 is a pass-through; every other phase is four products and a sum, rounded back to 16 bits (Fig. 5). A polynomial is a poor low-pass, so this core needs the cleanup filter.

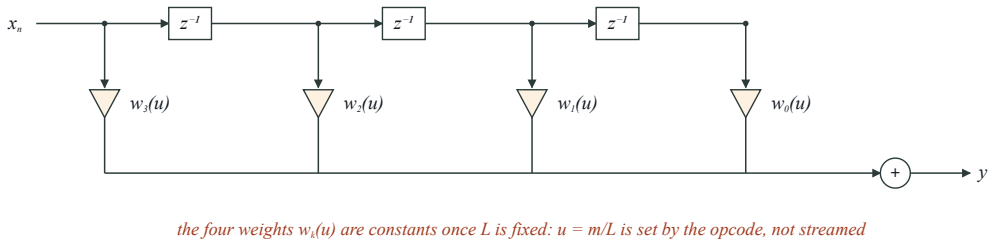


Figure 5. Cubic datapath. Four consecutive samples are weighted by $\mathbf{w}(u)$ and summed.

2.5 The FIR taps: the equiripple prototype

The polyphase prototype h of (6) is designed by the Parks–McClellan algorithm (MATLAB `firpm`). It is a minimax design: instead of the average error it minimises the worst-case error over the bands. With $D(f)$ the desired response, 1 in the passband and 0 in the stopband, and $W(f)$ a weight per band, the

weighted error is

$$E(f) = W(f)\,\big[H(f) - D(f)\big]. \tag{16}$$

The alternation theorem says that a linear-phase filter of a given length is optimal in this sense exactly when $|E(f)|$ reaches its maximum δ at a set of extremal frequencies f_i with alternating sign,

$$E(f_i) = (-1)^i\,\delta, \tag{17}$$

which is why the result is called equiripple: the error is spread into ripples of equal height rather than one large bump. The ripples in the two bands follow the weights,

$$\delta_{\text{pass}} = \frac{\delta}{W_{\text{pass}}}, \qquad \delta_{\text{stop}} = \frac{\delta}{W_{\text{stop}}}. \tag{18}$$

The Remez exchange finds the solution by iteration: guess the extremal frequencies on a dense grid; solve for the filter whose error is exactly $\pm\delta$ at those points; evaluate the error over the whole grid and exchange the guessed points for the points where the error is now largest; repeat until δ stops changing. The frequency samples are then turned back into the taps $h[k]$ by an inverse discrete Fourier transform.

The chips use the order $LP - 1$ ($7L$ taps), the bands $[0,\, f_p]$ and $[f_{st},\, Lf_s/2]$ and the weights $[80, 1000]$, so by (18) the stopband ripple is $12.5\times$ smaller than the passband ripple: image rejection is bought first, and the passband flatness that remains is enough for the EVM. The band edges are chosen per L because the image that matters moves with L:

$$(f_p, f_{st}) = \begin{cases} (14.0,\ 47.0)\ \text{MHz} & L \le 2, \\ (14.0,\ 48.5)\ \text{MHz} & L = 3, \\ (12.0,\ 48.5)\ \text{MHz} & L \ge 4. \end{cases} \tag{19}$$

At $L = 4$ this bought 7.7 dB of image rejection with EVM improved at the same time, at zero hardware cost. Taps are 16-bit, full scale $2^{15} - 1$.

2.6 Cleanup filter, fixed point and the substitution

The cubic and spline cores are followed by a 10-tap equiripple low-pass at the output rate, designed per core and per factor; symmetry is forced before quantising so that the RTL folds ten products into five. Taps are Q15 and products 32-bit; every core rounds its result back to 16 bits, and each core carries its own accumulator bound (Table 2).

Table 2. Accumulator width of each core, as declared in the RTL.

block	bits	what the accumulator holds
polyphase, Chip 1 (CFG)	36	the sum of seven products, loadable taps
polyphase, Chip 2 (NCFG)	32	the sum of seven products, hardwired taps
cubic	34	the sum of four products
spline	44	the three chained Horner stages
cleanup	36	the sum of five paired products

Chip 2 (NCFG) applies (2) to all 98 polyphase taps, at most five terms each, and to the cleanup taps in canonical-signed-digit form, at most three; neither file contains a multiplication operator (Fig. 6). In

the cubic and spline cores the substitution is used where the fractional delay is a power of two, at $L = 2$ and $L = 4$; at $L = 3$ and $L = 5$ those two cores keep a multiplier.

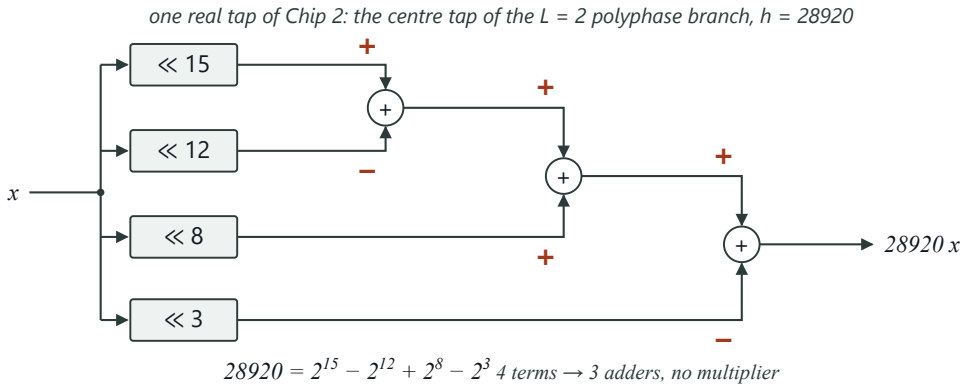


Figure 6. Signed power-of-two substitution on one real tap of Chip 2 (NCFG): the centre tap of the $L = 2$ polyphase branch is $28920 = 2^{15} - 2^{12} + 2^8 - 2^3$, so multiplying a sample by it is four shifts and three adders and no multiplier. The same is done for every one of the 98 polyphase taps and the cleanup taps.

Chapter 3

Architecture of the two chips

Both dies carry the same three cores behind one control unit and one cleanup filter (Figures 7 and 8). Chip 1 (CFG) loads every tap over a coefficient bus; Chip 2 (NCFG) hardwires the taps and has no bus. Everything else is identical, which makes the two chips a controlled comparison. The chip has two serial inputs, a clock, an asynchronous reset and 33 outputs: two 16-bit converter words and `dac_valid`. A frame begins with an 8-bit opcode, start(1) | core(2) | mode(2) | $L(3)$; an illegal opcode is rejected and the previous configuration stands. The control unit deserialises, decodes, generates the output strobe by dividing the clock by WL and counting L beats per word, and enables only the selected core. Delay-line enables are banks of distinct registers; the reset is seen only by a two-flop synchroniser. Every output leaves as flop $\rightarrow$ pad, so `dac_valid` is aligned with its data by construction. One clock domain at up to 960 MHz; clock gating with integrated cells only, 78 in Chip 1 (CFG) and 34 in Chip 2 (NCFG).

Clocks and word rates of the blocks. There is one clock, `clk`, at WL $\times$ 60 MHz: 720, 900 or 960 MHz for the 12-, 15- and 16-bit word lengths. Every register in the chip is clocked by it, but the blocks do their work at three different rates, and the timing constraints follow those rates (Table 3). The control unit works every clock, one serial bit per edge, and completes a word every WL clocks. The cores work once per input word, so their registers are enabled one clock in every WL, a word rate of 60 MHz; the paths between their word registers therefore have WL clocks to settle and are constrained as 13-cycle setup and 12-cycle hold multicycle paths, one cycle inside the shortest word of 12 clocks, while their delay-line recirculation loops, which shift every clock, stay single-cycle. The cleanup filter and the output stage work once per output word, every WL$/L$ clocks, a word rate of $L \times 60$ MHz from 120 to 300 MHz; the handoff from the cleanup filter into the output register is timed as the single cycle it is, and the 33 outputs are timed through the 2.1 ns pad as 4-cycle setup and 3-cycle hold multicycle paths, data and `dac_valid` together. The two serial inputs are single-cycle paths with an input delay of 0.75 to 0.95 ns after the pin edge, as the laboratory generator drives them. The mode registers of the control unit, which are static for a whole frame, are given 16 cycles setup and 15 hold into the cores.

How an unused core is switched off. The opcode selects one core, and the control unit derives one enable per core from that selection, the enable of the selected core following the word strobe and the other two held low. Synthesis turned every such enable into an integrated clock-gating cell on the clock of the registers it drives, so the registers of the two unselected cores receive no clock edges at all for the whole frame; they hold their state, and their only consumption is leakage and the clock-gate cells themselves. This is what the per-block power of Chapter 5 shows: the two idle cores together draw about 2 mW on either chip while the selected one draws the full figure.

Table 3. Clock, word rate and timing constraint of every block. One clock for the whole chip; the constraints are the sign-off SDC. DESIGN-PARAM.

block	clock	works every	timing constraint
reset synchroniser	`clk`	clock	single-cycle; asynchronous assert, synchronous release
control unit	`clk`	clock, one serial bit	single-cycle; inputs 0.75 to 0.95 ns after the pin; mode registers to the cores 16 setup / 15 hold
polyphase, cubic, spline	`clk`, enabled	input word, WL clocks	word registers 13 setup / 12 hold; recirculation loops single-cycle; core to cleanup 2 setup / 1 hold
cleanup filter	`clk`, enabled	output word, WL/L clocks	handoff to the output register single-cycle
output stage	`clk`	output word, WL/L clocks	registers to pads 4 setup / 3 hold, 0.1 ns output delay, `dac_valid` included
all	`clk`, 1.041 7 ns		uncertainty 0.13 setup / 0.05 hold; clock-gating check 0.3 setup / 0.1 hold; maximum fanout 20

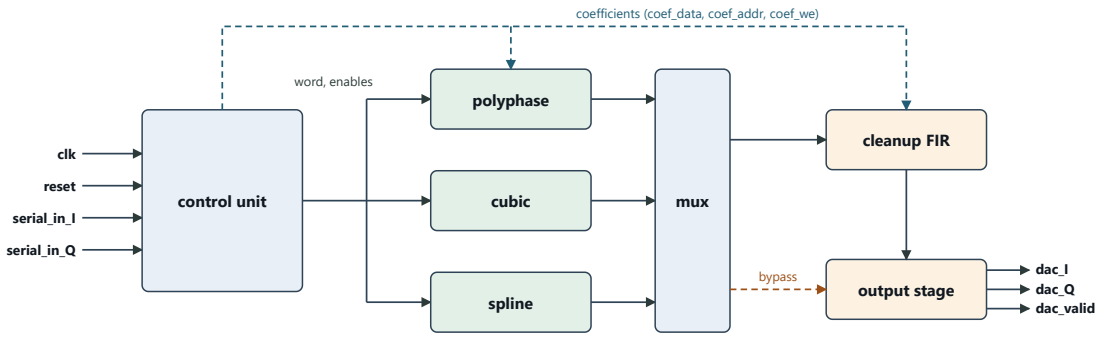


Figure 7. Chip 1 (CFG) (CFG). Three interpolator cores with runtime-loadable taps, fed by the control unit and a coefficient bus; the cleanup FIR serves the cubic and spline cores while the polyphase core bypasses it.

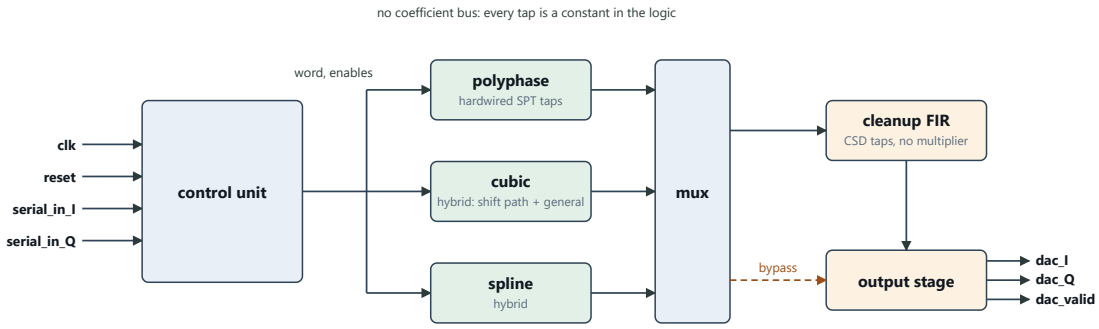


Figure 8. Chip 2 (NCFG) (NCFG). The same topology with every tap a constant in the logic; no coefficient bus.

Chapter 4

Design flow and verification method

Fig. 9 names the tool behind every stage: MATLAB for the golden model and the stimulus, VCS and Verdi for every simulation, VC Formal for the block proofs, Design Compiler for synthesis, Conformal for equivalence, Innovus for the physical design, Voltus for power, PrimeTime for sign-off timing, Calibre for the physical checks, and the Keysight analyser on the bench. The flow is a fixed sequence of representations of one design, and one golden model that every representation is compared against (Fig. 10). One script does every compare: it counts output lines and unknown values, aligns by the pipeline offset and reports per core and per segment whether every word matched. Every output must be newer than the netlist it came from, and every netlist carries an identity, its clock-gate and flip-flop counts, that must match (Fig. 12). Fig. 11 shows how one function travels from the floating-point model through the fixed-point model to the SystemVerilog.

4.1 Technology

TSMC 65 nm LP, nine metals. All three threshold flavours of the LP standard-cell library are offered to synthesis and to place-and-route together: `tcbn65lp` (standard threshold), `tcbn65lplvt` (low threshold, for the fast paths) and `tcbn65lphvt` (high threshold, for leakage), so the tools pick the flavour per cell. Pads from `tpdn65lpnv2od3`. Synthesis maps at 1.08 V, 125 °C; place-and-route and sign-off time a slow and a fast view, setup and hold in both, hold signed at the fast corner.

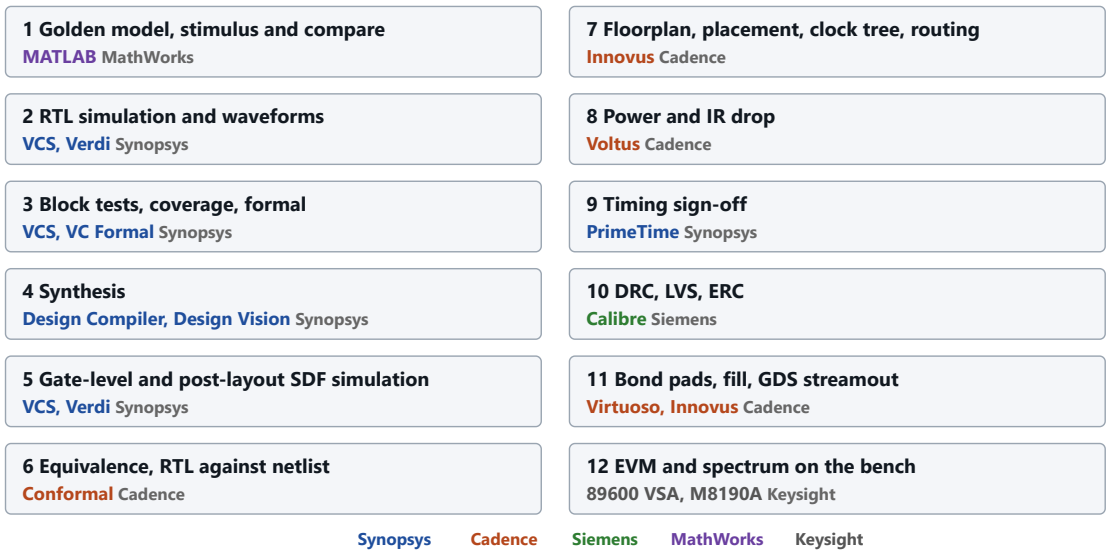


Figure 9. The tool used at each stage, from the MATLAB model to the bench.

4.2 The stages and their gates

Fig. 10 lists the stages in order, with what each one produces and the gate that admits the next. The rest of this section walks through them one by one, each with the picture it produced.

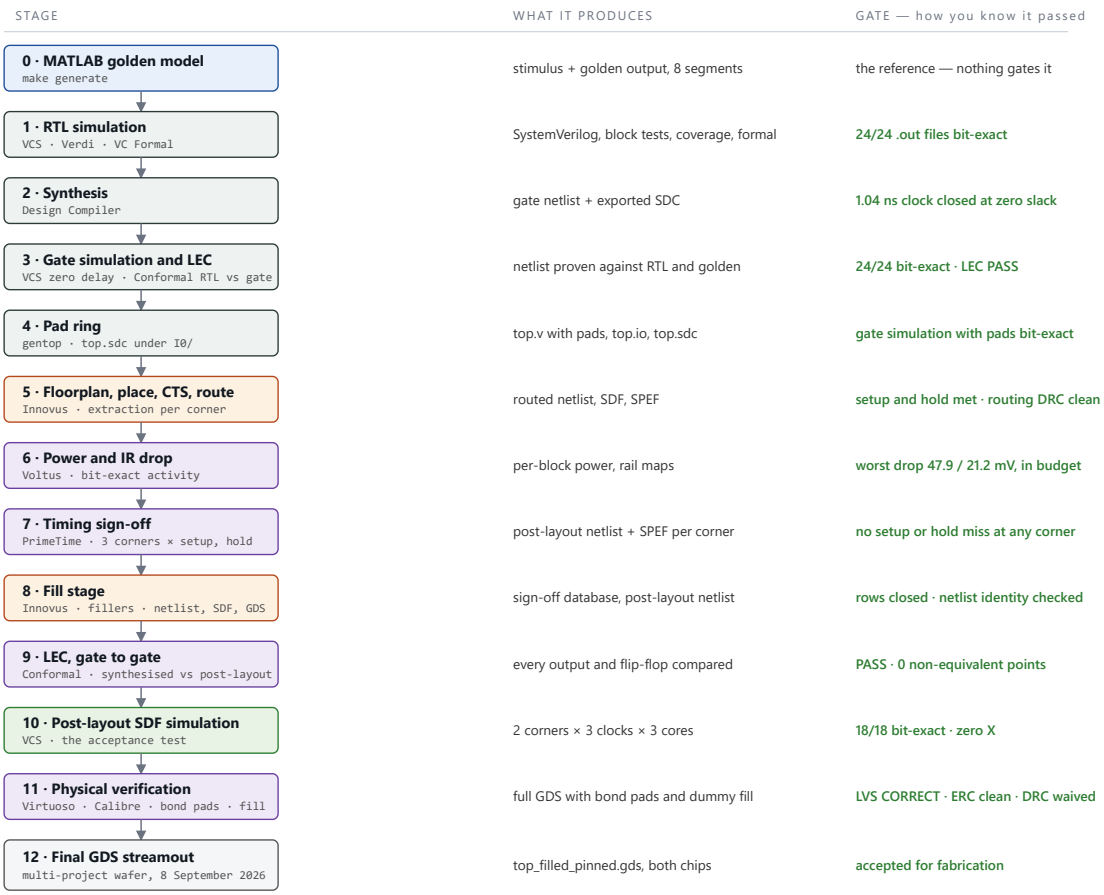


Figure 10. The flow and its gates. Every stage is admitted only when its output matches the golden sample for sample; the post-layout simulation of the routed netlist is the acceptance test.

4.2.1 Golden model and stimulus

MATLAB generates the QAM-64 stream, the filter coefficients and the expected output words of every legal mode, eight segments per core. The same fixed-point model that wrote the taps writes the golden files, so the reference is the arithmetic itself, not a second implementation of it. Fig. 11 shows how one function travels from the floating-point model through the fixed-point model to the SystemVerilog, and Fig. 12 the one stream that feeds both the model and every simulation.



```
function y_up = interp_minAJ2(x, L)
% MinAJ2 cubic Hermite interpolation
% x : input samples (real or complex)
% L : upsampling factor

x = x(:);
N = length(x);
Nout = (N-1)*L + 1;

% ---- CHANGE IS HERE ----
y_up   = complex(zeros(Nout,1));   % instead of zeros(Nout,1)
% ------------------------

outIdx = 1;

% ----------------------------------
% Hermite basis
% ----------------------------------
u  = (0:L-1).' / L;
u2 = u .* u;
u3 = u2 .* u;

H00 =  2*u3 - 3*u2 + 1;
H01 = -2*u3 + 3*u2;
H10 =  u3 - 2*u2 + u;
H11 =  u3 - u2;

% ----------------------------------
% Startup slopes
% ----------------------------------
s0 = x(2) - x(1);
s1 = 0.5*(x(3) - x(1));
```

(a) floating-point MATLAB

```
function y_up = minAJ2_fixed(x, L, wl, fl)
% MinAJ2 cubic Hermite interpolation (fixed-point)
% x : input samples (real or complex)
% L : upsampling factor

    % Final output type
    T_sig = numerictype(1, wl, fl);
    F_sig = fimath('RoundingMethod','Nearest', ...
                   'OverflowAction','Saturate', ...
                   'ProductMode','SpecifyPrecision', ...
                   'ProductWordLength', wl, ...
                   'ProductFractionLength', fl, ...
                   'SumMode','SpecifyPrecision', ...
                   'SumWordLength', wl, ...
                   'SumFractionLength', fl);

    wlA = wl + 3;
    flA = fl + 2;
    T = numerictype(1, wlA, flA);
    F = fimath( ...
        'RoundingMethod','Nearest', ...
        'OverflowAction','Saturate', ...
        'ProductMode','SpecifyPrecision', ...
        'ProductWordLength', wlA+flA, ...
        'ProductFractionLength', flA+flA, ...
        'SumMode','SpecifyPrecision', ...
        'SumWordLength', wlA, ...
        'SumFractionLength', flA);

    % constant coefficients
    one   = fi( 1,   T, F);
    two   = fi( 2,   T, F);
    m2    = fi(-2,   T, F);
    three = fi( 3,   T, F);
    m3    = fi(-3,   T, F);
```

(b) fixed-point MATLAB

```
module minaj2_interp_3samp_internalSlope (
  input  logic                clk,
  input  logic                reset,
  input  logic                en,
  input  logic                strobe,
  input  logic [2:0]          L_ctrl,
  input  logic signed [15:0] x_prev,
  input  logic signed [15:0] x_c,
  input  logic signed [15:0] x_n,
  output logic signed [15:0] y0, y1, y2, y3, y4
);


  logic signed [15:0] m_p;
  logic signed [15:0] m_new;


  logic signed [17:0] c0, c1, c2, c3;


  logic signed [21:0] term_11_xp;
  logic signed [21:0] term_4_mp;
  logic signed [21:0] term_8_xc;
  logic signed [21:0] term_3_xn;
  logic signed [21:0] Sum;

  logic signed [36:0] mult_res;
  logic signed [23:0] raw_m;

  logic signed [16:0] delta;
  logic signed [18:0] term_3delta;
  logic signed [18:0] term_2m0;
  logic signed [18:0] term_2delta;
```

(c) SystemVerilog

Figure 11. From the model to the RTL, shown on the spline core's slope recursion: the floating-point function, the same function with the word lengths, rounding and saturation fixed, and the SystemVerilog module with the same widths.

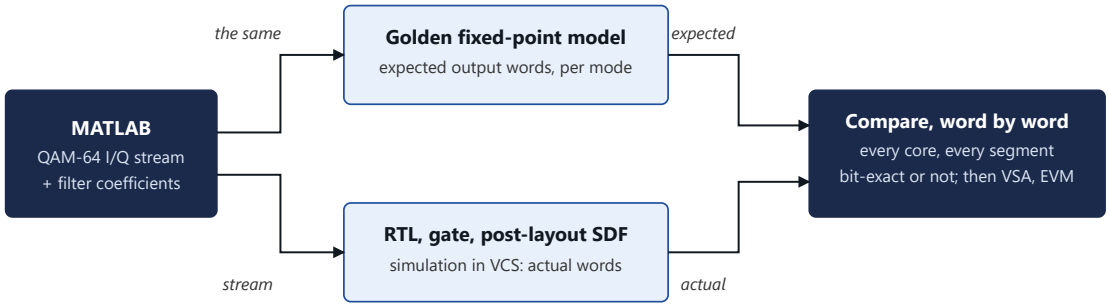


Figure 12. One stream, two paths. MATLAB produces the QAM-64 stream and the coefficients; the golden fixed-point model turns them into the expected words, the simulated hardware into the actual words, and the compare reports per core and per segment whether every word matched.

4.2.2 RTL simulation

Every core and every segment is simulated in VCS and compared word by word: 8 of 8 segments bit-exact on each core, both chips (Fig. 13). Each block also carries its own directed test with assertions, functional coverage groups and formal proofs in VC Formal (Fig. 14).

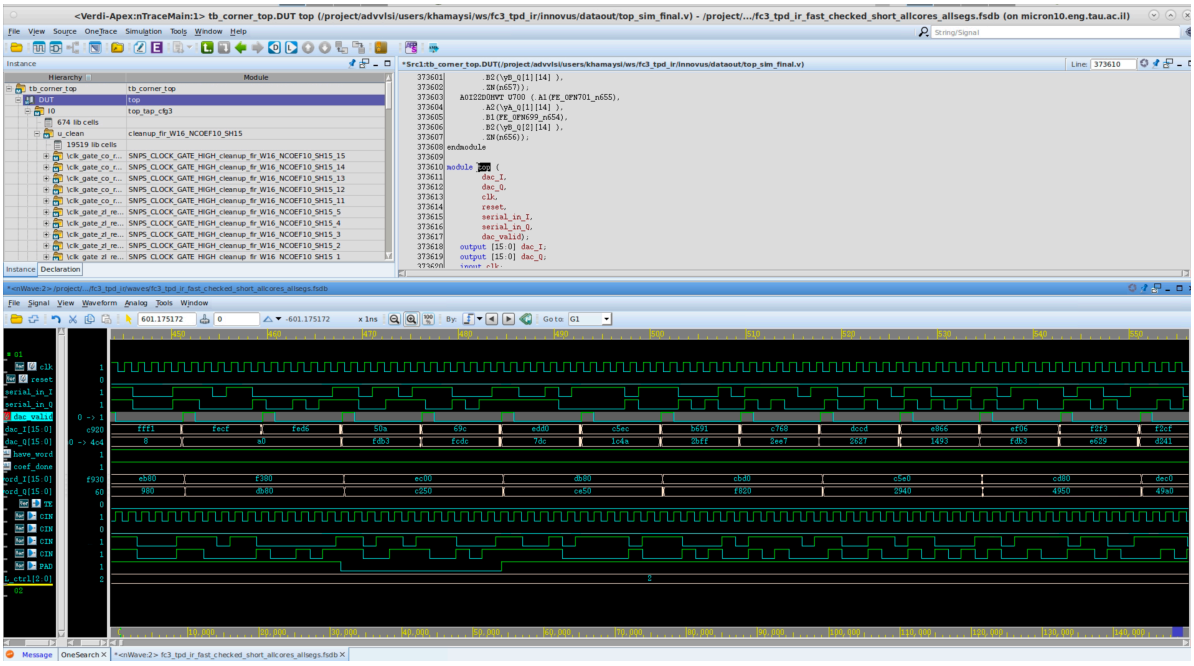


Figure 13. Bit-exact output on the waveform at 960 MHz, all cores and all segments: `dac_valid` frames every output word and the converter bus carries the golden values.

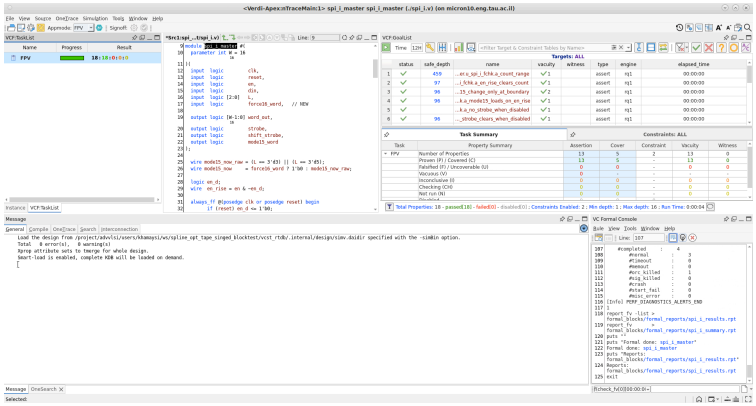
```
============================================================
 TB: spi_cfg4_nomode - posedge-synchronous TB with SVA
============================================================

--- TEST 1: Reset state ---
  PASS  [reset/L_ctrl] got=2 exp=2
  PASS  [reset/cfg_done] got=0 exp=0
  PASS  [reset/cfg_ready] got=0 exp=0

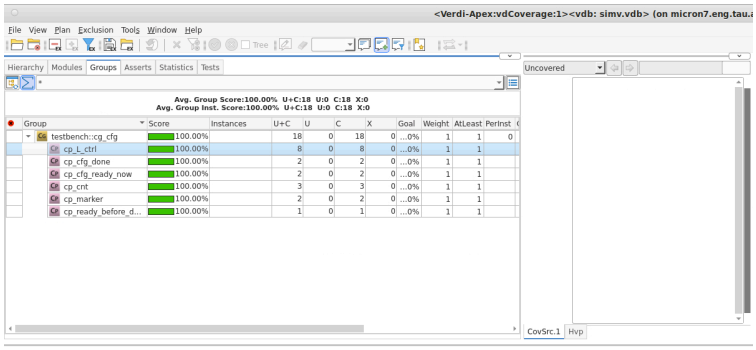
--- TEST 2: Valid L encodings ---
  PASS  [L=2/L_ctrl] got=2 exp=2
  PASS  [L=2/cfg_done] got=1 exp=1
  PASS  [L=2/cfg_ready] got=1 exp=1
  PASS  [L=3/L_ctrl] got=3 exp=3
  PASS  [L=3/cfg_done] got=1 exp=1
  PASS  [L=3/cfg_ready] got=1 exp=1
  PASS  [L=4/L_ctrl] got=4 exp=4
  PASS  [L=4/cfg_done] got=1 exp=1
  PASS  [L=4/cfg_ready] got=1 exp=1
  PASS  [L=5/L_ctrl] got=5 exp=5
  PASS  [L=5/cfg_done] got=1 exp=1
  PASS  [L=5/cfg_ready] got=1 exp=1

--- TEST 3: cfg_ready_now timing ---
  PASS  [before_header/cfg_ready_now] got=0 exp=0
  PASS  [before_header/cfg_done] got=0 exp=0
  PASS  [after_marker/cfg_ready_now] got=0 exp=0
  PASS  [after_marker/cfg_done] got=0 exp=0
  PASS  [after_L0/cfg_ready_now] got=0 exp=0
  PASS  [after_L0/cfg_done] got=0 exp=0
  PASS  [after_L1/cfg_ready_now] got=1 exp=1
  PASS  [after_L1/cfg_done] got=0 exp=0
  PASS  [after_L2/cfg_ready_now] got=1 exp=1
  PASS  [after_L2/cfg_done] got=1 exp=1
  PASS  [after_L2/L_ctrl] got=2 exp=2
```

(a) directed test with assertions



(b) functional coverage in Verdi



(c) formal proofs in VC Formal

Figure 14. Block-level verification of the serial configuration block, from the earlier iteration of the design: (a) a directed SystemVerilog test with assertions, every check PASS; (b) its functional coverage groups at 100 %; (c) VC Formal proving the block's 18 properties with none falsified.

4.2.3 Synthesis

Design Compiler maps the RTL onto the three threshold flavours of the library with the hierarchy kept and integrated clock gates only. Both chips close the 1.04 ns clock at zero slack: 54 665 cells and 40 logic levels on Chip 1 (CFG), 38 749 cells and 33 levels on Chip 2 (NCFG). Figures 15 to 18 show the synthesised netlists in Design Vision, from the hierarchy down to the gates of one block.

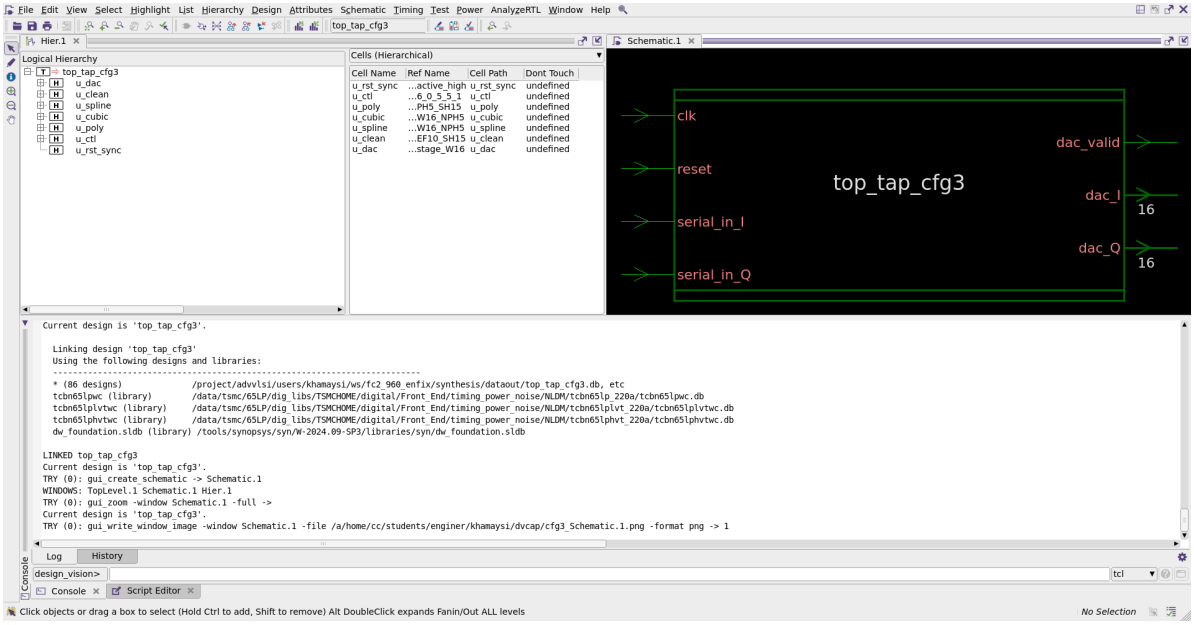


Figure 15. The synthesised netlist of Chip 1 (CFG) opened in Design Vision: the logical hierarchy under `top_tap_cfg3`, the seven hierarchical cells with their synthesised reference names, and the block schematic drawn from the gate netlist.

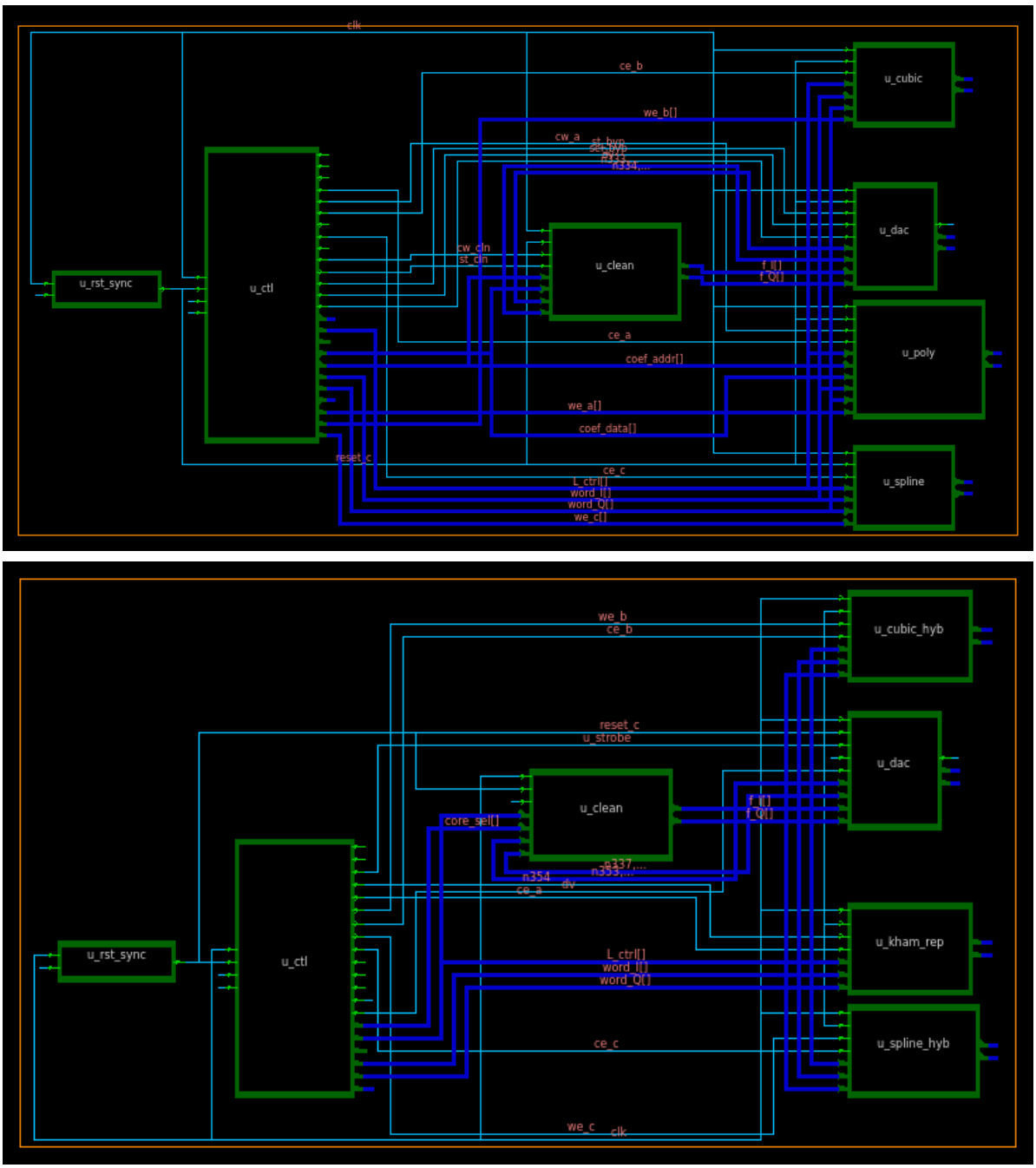
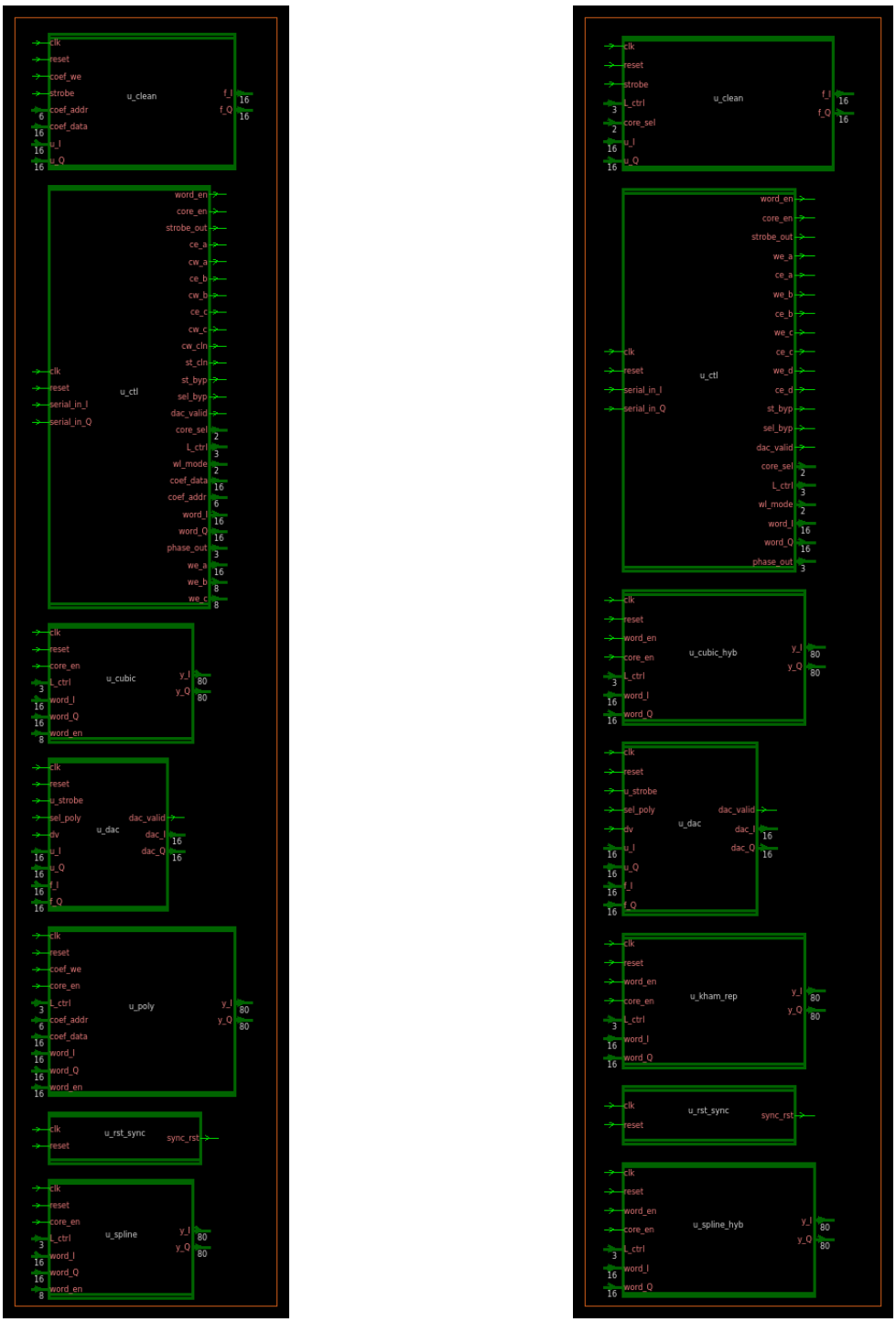
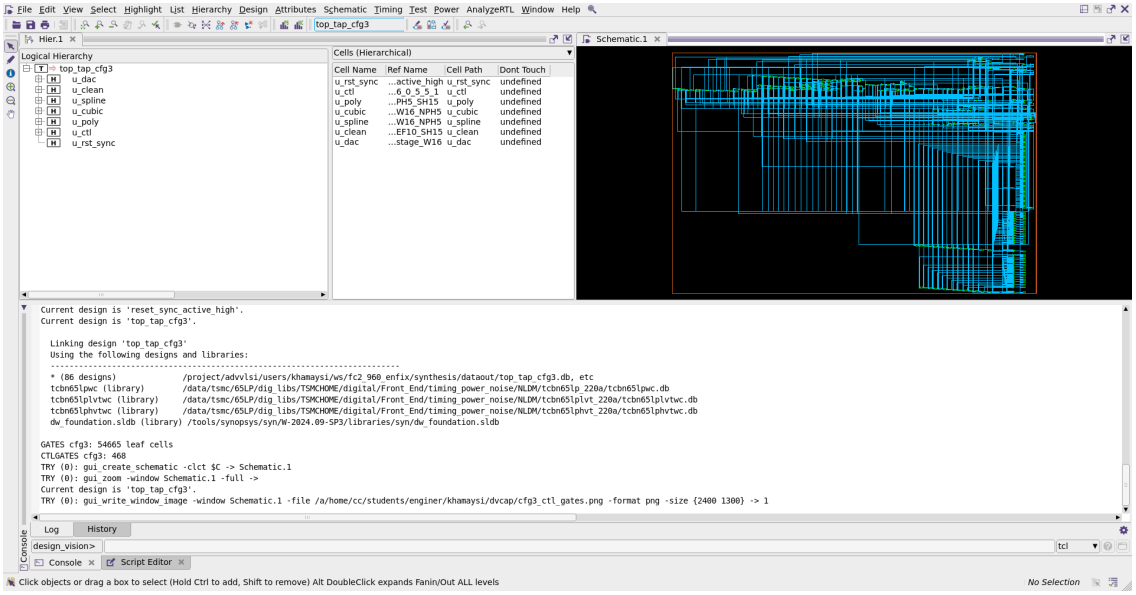


Figure 16. The blocks after synthesis, drawn by Design Vision from the gate netlists: the reset synchroniser, the control unit, the three cores, the cleanup filter and the output stage with the nets between them. Top: Chip 1 (CFG), with the coefficient bus (`coef_data`, `coef_addr`, `we_a/b/c`) from the control unit into the cores. Bottom: Chip 2 (NCFG), the same topology without the bus. The stubs on the right edge of each core are its result buses, 80 bits each, the five phases of 16 bits. They do not go to another block but to the output selector, about 340 standard cells that synthesis left at the top level, which picks the active core and the current phase and feeds the 16-bit word into the cleanup filter for the cubic and spline, or straight into the output stage for the polyphase, the `u_I` and `u_Q` inputs of those two blocks. Design Vision draws only block-to-block nets in this view, so that path is not visible here; Fig. 7 shows it. Only the three pins on the right of the output stage, `dac_I`, `dac_Q` and `dac_valid`, leave the chip.

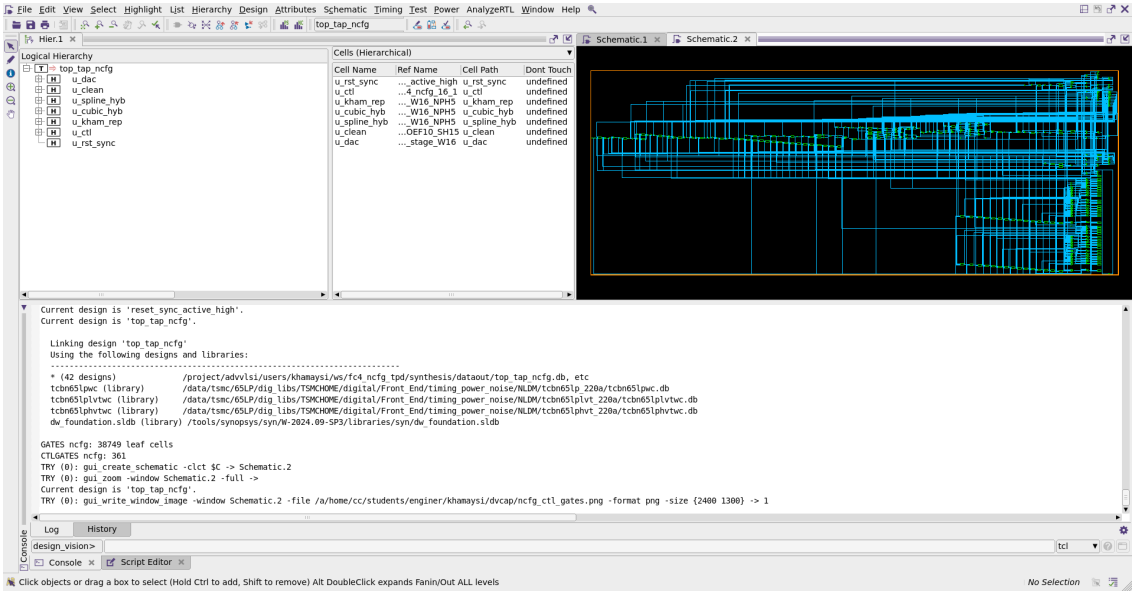


(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 17. The same seven synthesised blocks drawn one under the other with every port and bus width, from Design Vision. The control unit's outputs are the per-core enables, the word and coefficient buses and the mode registers; each core returns an 80-bit bus of five 16-bit phase outputs; the cleanup filter and the output stage carry 16-bit I and Q. The hardwired chip's cores have no coefficient inputs.

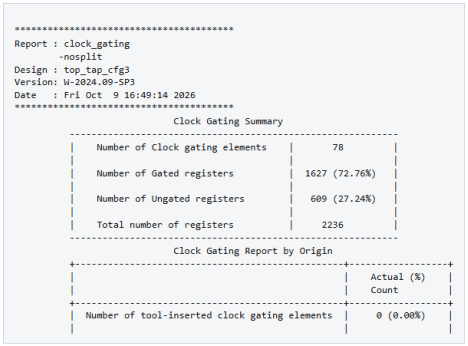


(a) Chip 1 (CFG)



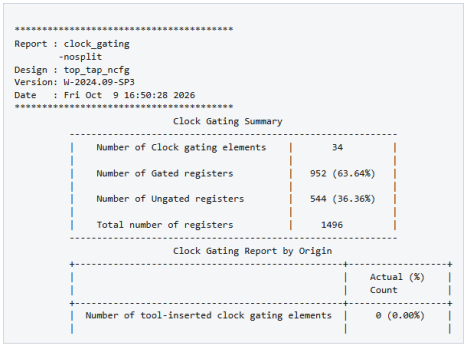
(b) Chip 2 (NCFG)

Figure 18. The control unit after synthesis, drawn at gate level in Design Vision from the netlist that went to place-and-route: every standard cell and every wire of the block.



```
****************************************
Report : clock_gating
         -nosplit
Design : top_tap_cfg3
Version: W-2024.09-SP3
Date   : Fri Oct  9 16:49:14 2026
****************************************
                          Clock Gating Summary
  ------------------------------------------------------------
  |    Number of Clock gating elements     |       78         |
  |                                        |                  |
  |    Number of Gated registers           |  1627 (72.76%)   |
  |                                        |                  |
  |    Number of Ungated registers         |   609 (27.24%)   |
  |                                        |                  |
  |    Total number of registers           |     2236         |
  ------------------------------------------------------------
                          Clock Gating Report by Origin
  +-----------------------------------------+-----------------+
  |                                         |    Actual (%)   |
  |                                         |    Count        |
  +-----------------------------------------+-----------------+
  |  Number of tool-inserted clock gating elements  |     0 (0.00%)   |
  |                                         |                 |
```

(a) Chip 1 (CFG): 78 gates, 1 627 of 2 236 registers gated



```
****************************************
Report : clock_gating
         -nosplit
Design : top_tap_ncfg
Version: W-2024.09-SP3
Date   : Fri Oct  9 16:50:28 2026
****************************************
                          Clock Gating Summary
  ------------------------------------------------------------
  |    Number of Clock gating elements     |       34         |
  |                                        |                  |
  |    Number of Gated registers           |   952 (63.64%)   |
  |                                        |                  |
  |    Number of Ungated registers         |   544 (36.36%)   |
  |                                        |                  |
  |    Total number of registers           |     1496         |
  ------------------------------------------------------------
                          Clock Gating Report by Origin
  +-----------------------------------------+-----------------+
  |                                         |    Actual (%)   |
  |                                         |    Count        |
  +-----------------------------------------+-----------------+
  |  Number of tool-inserted clock gating elements  |     0 (0.00%)   |
  |                                         |                 |
```

(b) Chip 2 (NCFG): 34 gates, 952 of 1 496 registers gated

Figure 19. The clock-gating report of each synthesised netlist, from Design Compiler. Every gating cell is one the RTL asked for, through the enables of the control unit, none was inserted by the tool. On Chip 1 (CFG) the 78 cells are 42 in the polyphase core, 16 in the cleanup filter, 9 in the control unit, 5 each in the cubic and spline cores and 1 in the output stage; on Chip 2 (NCFG) the 34 are 11 in the cleanup filter, 7 in the control unit, 5 in each of the three cores and 1 in the output stage. The ungated registers are the serial front end and the delay lines that shift every clock.

4.2.4 Zero-delay gate-level simulation

The gate netlist is simulated without delays against the same golden files: 24 of 24 segment–core pairs bit-exact, including the two illegal headers that must be rejected (Fig. 20).

```
GLS0-GATE: 24/24 data segments bit-exact vs golden_out_c2/; 30/30 files byte-identical (expect 30/30)
=== seg01_WL12_L2.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     19264    19212       0     19264  BIT-EXACT (of 19264)
cubic_hyb    19264    19214       0     19264  BIT-EXACT (of 19264)
spline_hyb   19264    19228       0     19264  BIT-EXACT (of 19264)
=== seg02_WL12_L3.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     28864    28818       0     28864  BIT-EXACT (of 28864)
cubic_hyb    28864    28817       0     28864  BIT-EXACT (of 28864)
spline_hyb   28864    28837       0     28864  BIT-EXACT (of 28864)
=== seg03_WL12_L4.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     38464    38424       0     38464  BIT-EXACT (of 38464)
cubic_hyb    38464    38419       0     38464  BIT-EXACT (of 38464)
spline_hyb   38464    38449       0     38464  BIT-EXACT (of 38464)
=== seg04_WL15_L3.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     28864    28818       0     28864  BIT-EXACT (of 28864)
cubic_hyb    28864    28817       0     28864  BIT-EXACT (of 28864)
spline_hyb   28864    28838       0     28864  BIT-EXACT (of 28864)
=== seg05_WL15_L5.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     48064    48030       0     48064  BIT-EXACT (of 48064)
cubic_hyb    48064    48023       0     48064  BIT-EXACT (of 48064)
spline_hyb   48064    48060       0     48064  BIT-EXACT (of 48064)
=== seg06_WL16_L2.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     19264    19212       0     19264  BIT-EXACT (of 19264)
cubic_hyb    19264    19214       0     19264  BIT-EXACT (of 19264)
spline_hyb   19264    19226       0     19264  BIT-EXACT (of 19264)
=== seg07_WL16_L4.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     38464    38424       0     38464  BIT-EXACT (of 38464)
cubic_hyb    38464    38419       0     38464  BIT-EXACT (of 38464)
spline_hyb   38464    38446       0     38464  BIT-EXACT (of 38464)
=== seg08_WL16_L4.txt ===
core         lines  nonzero  offset   matched  verdict
kham_rep     16448    16400       0     16448  BIT-EXACT (of 16448)
cubic_hyb    16448    16395       0     16448  BIT-EXACT (of 16448)
spline_hyb   16448    16428       0     16448  BIT-EXACT (of 16448)
  cmp_seg.py (GLS/out vs out/ RTL) rc=0
GLS0-GATE PASS
GLS0-DONE 2026-09-07 15:29:48
```

Figure 20. The compare verdict of the zero-delay gate-level simulation of Chip 2 (NCFG): every segment of every core bit-exact against the golden files, offset 0, no unknowns.

4.2.5 Equivalence check, RTL against the gate netlist

Before anything physical is done, Conformal proves the synthesised netlist equivalent to the RTL: PASS on both chips, the 33 outputs and the top-level registers compared directly and the six blocks proven as hierarchical black boxes (Fig. 21). The same check is repeated gate to gate after the fill stage (Section 4.2.15).

Conformal, RTL against the synthesised netlist: verification report

```
================================================================================
                          Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                      0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                 0
--------------------------------------------------------------------------------
3. User modification to design:                                             0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:  0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                        0
--------------------------------------------------------------------------------
6. Compare Results:                                                      PASS
================================================================================
```

Conformal, compared points (the six blocks proven as hierarchical black boxes)

```
43 compared point(s) reported
================================================================================
Compared points      PO    DFF    BBOX      Total
--------------------------------------------------------------------------------
Equivalent           33    4      6         43
```

(a) Chip 1 (CFG)

Conformal, RTL against the synthesised netlist: verification report

```
================================================================================
                          Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                      0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                 0
--------------------------------------------------------------------------------
3. User modification to design:                                             0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:  0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                        0
--------------------------------------------------------------------------------
6. Compare Results:                                                      PASS
================================================================================
```

Conformal, compared points (the six blocks proven as hierarchical black boxes)

```
43 compared point(s) reported
================================================================================
Compared points      PO    DFF    BBOX      Total
--------------------------------------------------------------------------------
Equivalent           33    4      6         43
```

(b) Chip 2 (NCFG)

Figure 21. Conformal, RTL against the synthesised netlist, both chips: compare result PASS, 43 compared points, all equivalent.

4.2.6 Pad ring

A script wraps the core in the pad ring, 37 signal and 9 supply pads with 4 mA drivers, writes the pad placement seed and reparents the constraints under the core instance (Fig. 22). The gate simulation is repeated through the pads and stays bit-exact.

```
gen_top.pl: wrote top.v and top.io
  ports: 4 input(s), 3 output(s); signal pads I5..I41
  gentop/top.sdc  (get_cells reparented under I0/ ; ports unchanged)
============================================================
gentop done:
  gentop/top.v   (pad-wrapped top, ready for GLS + Innovus)
  gentop/top.io  (pad placement seed - space pads evenly as needed)
  gentop/top.sdc (constraints reparented under I0/)
Next: 'make gentop_comp gentop_run' for the top-level gate sim,
      or read top.v into Innovus for floorplan/place-and-route.
============================================================
```

Figure 22. The pad-ring script's report for Chip 2 (NCFG): the core wrapped in signal pads I5 to I41, the pad placement seed written, and the constraints reparented under the core instance.

4.2.7 Timing views for place-and-route

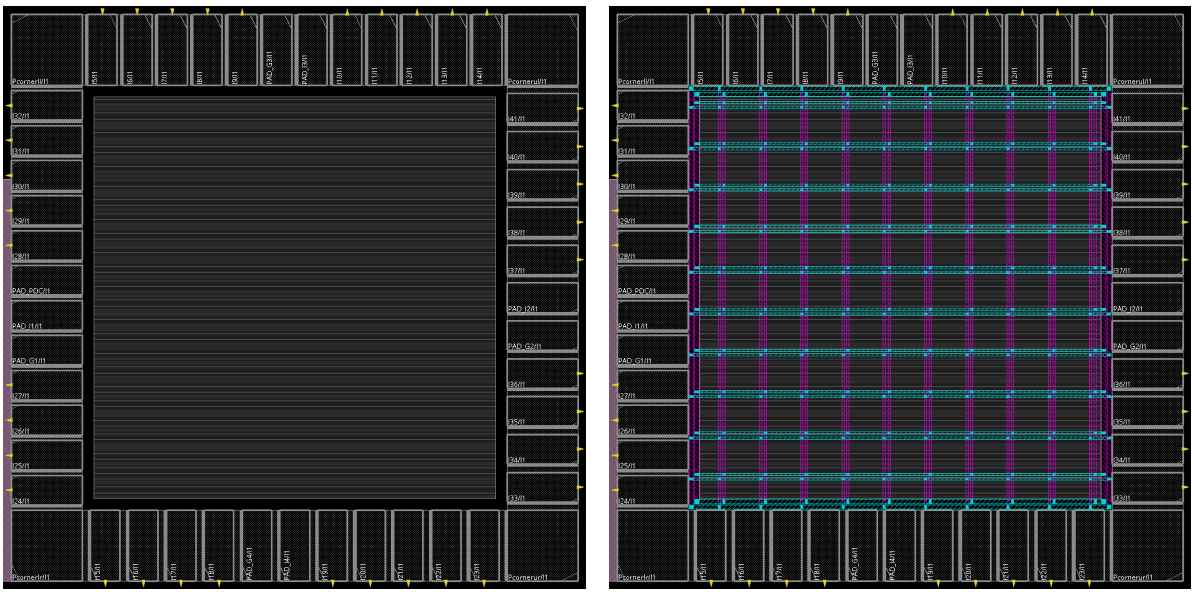
Before Innovus is started, the corners it must time are declared in one file, the multi-mode multi-corner view definition (Fig. 23). It names three RC corners from the TSMC capacitance tables, worst at 125 °C, best at −40 °C and typical at 25 °C; three library sets, the worst-, best- and typical-case timing libraries of the standard, low- and high-threshold cells and of the pads; one constraint mode, the SDC of the chip; and from these three delay corners and three analysis views. Setup is analysed in the slow and the fast view and hold in the fast and the slow view, so that no corner is left to a single mode; the typical view exists for the power analysis only.

```
create_rc_corner -name SlowRC  -cap_table $CAPTABLE/cln65lp_1p09m+alrdl_top2_cworst.captable -T {125} \
    -preRoute_res {1} -preRoute_cap {1} -preRoute_clkres {0} -preRoute_clkcap {0} \
    -postRoute_res {1} -postRoute_cap {1} -postRoute_xcap {1} \
    -postRoute_clkres {0} -postRoute_clkcap {0}
create_rc_corner -name FastRC  -cap_table $CAPTABLE/cln65lp_1p09m+alrdl_top2_cbest.captable -T {-40} \
    -preRoute_res {1} -preRoute_cap {1} -preRoute_clkres {0} -preRoute_clkcap {0} \
    -postRoute_res {1} -postRoute_cap {1} -postRoute_xcap {1} \
    -postRoute_clkres {0} -postRoute_clkcap {0}
create_rc_corner -name TypRC  -cap_table $CAPTABLE/cln65lp_1p09m+alrdl_top2_typical.captable -T {25} \
    -preRoute_res {1} -preRoute_cap {1} -preRoute_clkres {0} -preRoute_clkcap {0} \
    -postRoute_res {1} -postRoute_cap {1} -postRoute_xcap {1} \
    -postRoute_clkres {0} -postRoute_clkcap {0}
create_library_set -name Typ -timing {
    $NLDM/tcbn65lplvt_220a/tcbn65lplvttc.lib
    $NLDM/tcbn65lphvt_220a/tcbn65lphvttc.lib
    $NLDM/tcbn65lp_220a/tcbn65lptc.lib
    $NLDM/tpdn65lpnv2od3_200a/tpdn65lpnv2od3tc.lib
}
create_library_set -name Min -timing {
    $NLDM/tcbn65lplvt_220a/tcbn65lplvtbc.lib
    $NLDM/tcbn65lphvt_220a/tcbn65lphvtbc.lib
    $NLDM/tcbn65lp_220a/tcbn65lpbc.lib
    $NLDM/tpdn65lpnv2od3_200a/tpdn65lpnv2od3bc.lib
}
create_library_set -name Max -timing {
    $NLDM/tcbn65lplvt_220a/tcbn65lplvtwc.lib
    $NLDM/tcbn65lphvt_220a/tcbn65lphvtwc.lib
    $NLDM/tcbn65lp_220a/tcbn65lpwc.lib
    $NLDM/tpdn65lpnv2od3_200a/tpdn65lpnv2od3wc.lib
}
create_constraint_mode -name CM -sdc_files {../datain/top.sdc}
create_delay_corner -name DC_Slow -library_set {Max} -rc_corner {SlowRC}
create_delay_corner -name DC_Fast -library_set {Min} -rc_corner {FastRC}
create_delay_corner -name DC_Typ  -library_set {Typ} -rc_corner {TypRC}
create_analysis_view -name AV_Slow -constraint_mode {CM} -delay_corner {DC_Slow}
create_analysis_view -name AV_Fast -constraint_mode {CM} -delay_corner {DC_Fast}
create_analysis_view -name AV_Typ  -constraint_mode {CM} -delay_corner {DC_Typ}
set_analysis_view -setup {AV_Slow AV_Fast} -hold {AV_Fast AV_Slow}
```

Figure 23. The multi-mode multi-corner view definition of Chip 1 (CFG), as read by Innovus; Chip 2 (NCFG) uses the same file. The library directories are abbreviated to `$NLDM` and `$CAPTABLE`.

4.2.8 Floorplan and power grid

In Innovus: a 960×960 µm die, a 680×680 µm core, supply rings and five-finger straps from every supply pad (Fig. 24).

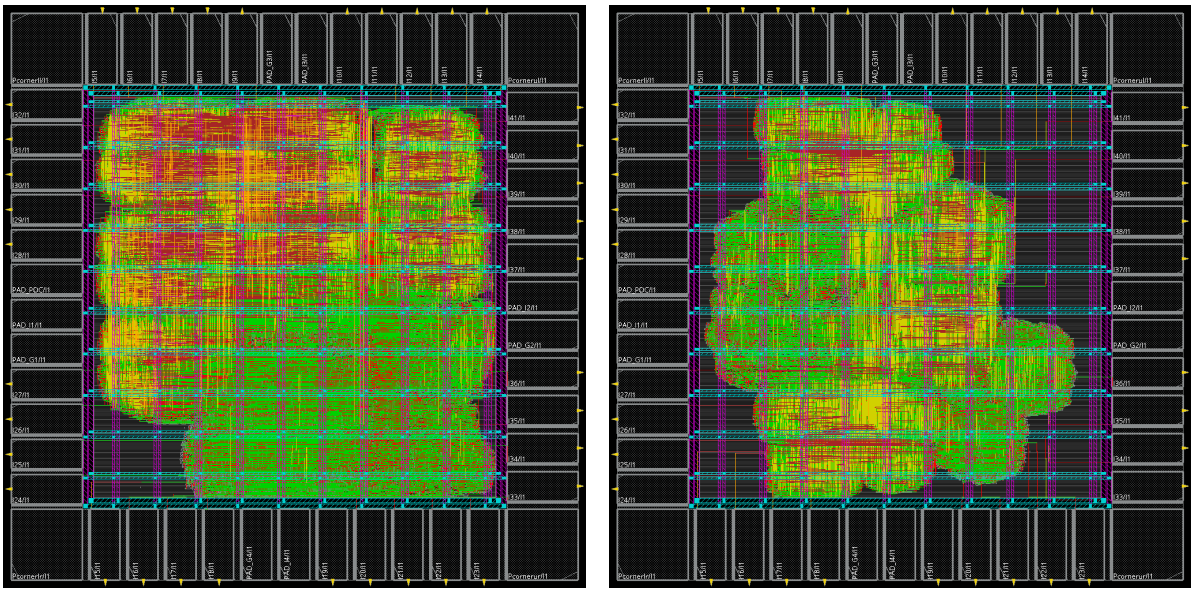


(a) floorplan database **(b)** power-plan database

Figure 24. Floorplan and power plan, captured from the stage databases: the pad ring and the empty core rows; the supply rings and the five-finger straps.

4.2.9 Placement

Density 80.13 % on Chip 1 (CFG) and 55.86 % on Chip 2 (NCFG), utilisation 64.4 and 51.2 % (Fig. 25).

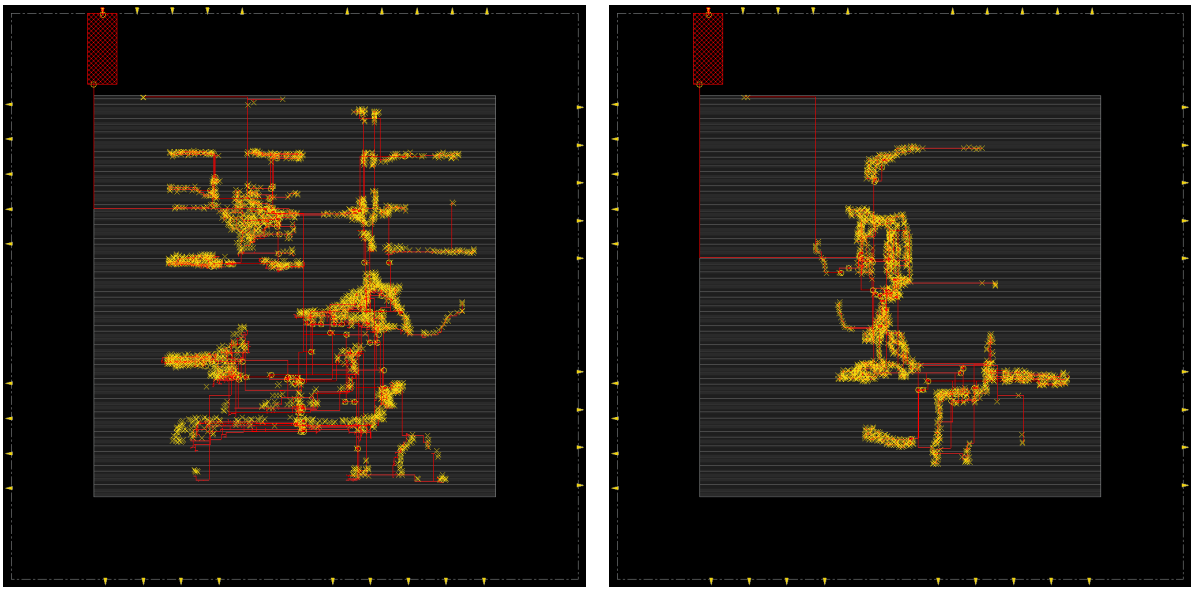


(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 25. Both chips after placement optimisation; the hardwired chip occupies visibly less of the same core.

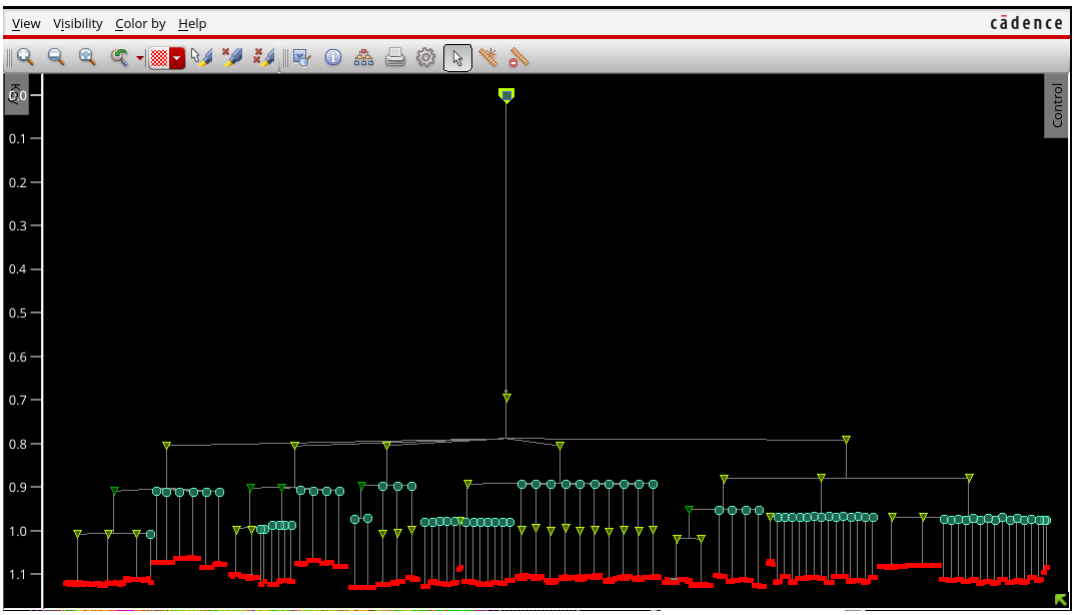
4.2.10 Clock tree

2 236 sinks on Chip 1 (CFG), balanced by CCOpt; the gate is the log line “Found 0 advancing pin insertion delay”. Fig. 26 shows the clock nets alone and Fig. 27 the trees in the Clock Tree Debugger.

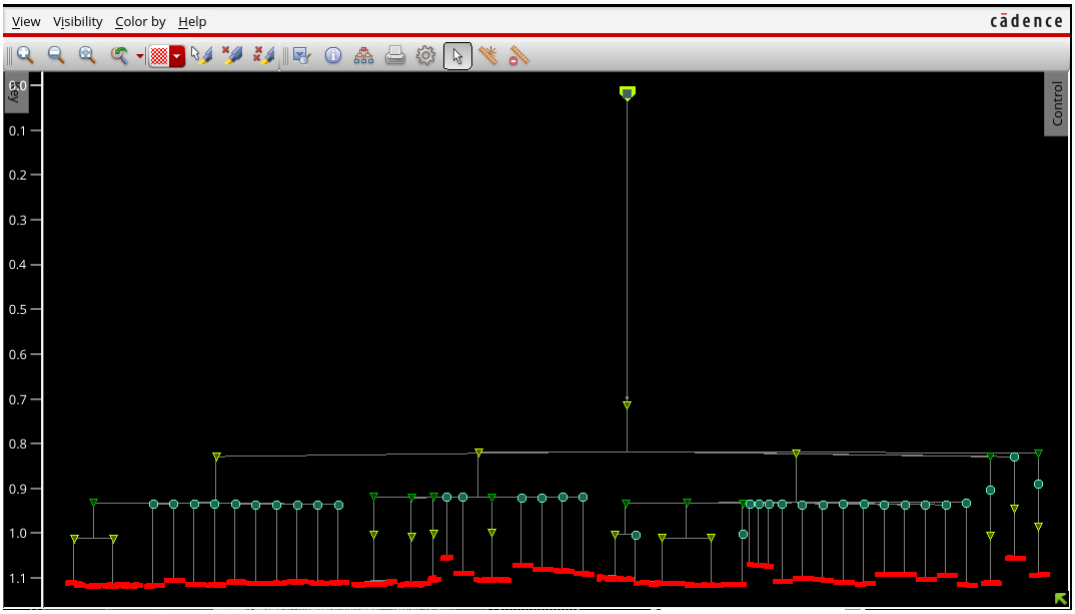


(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 26. The clock network of each chip alone: the nets reached from the clock pad through the buffers and integrated gates, in red.



(a) Chip 1 (CFG)

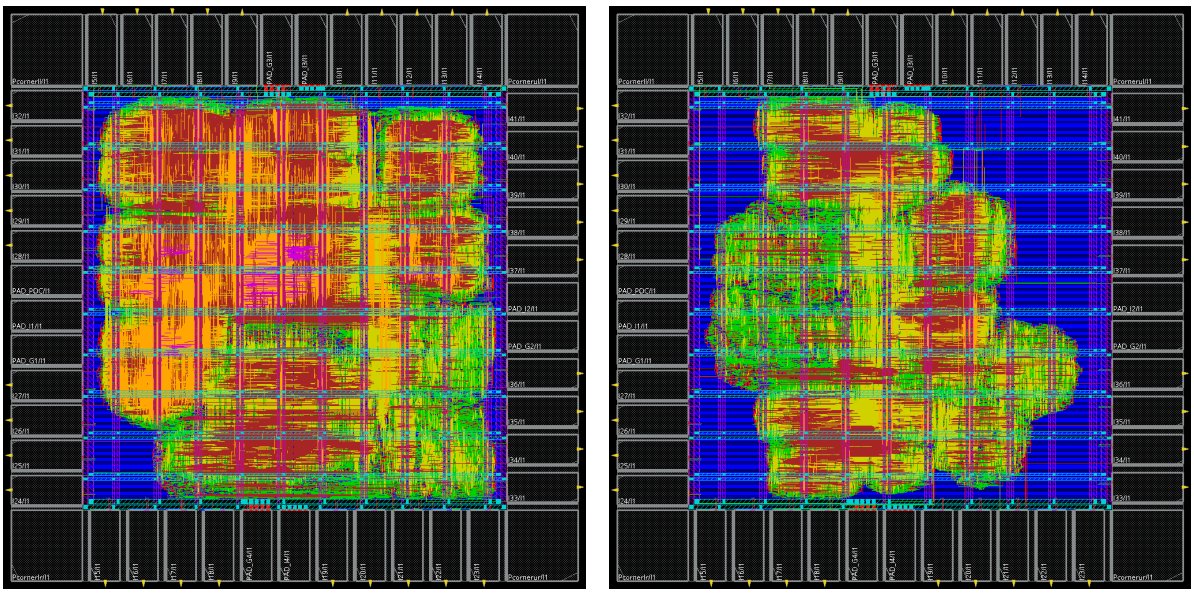


(b) Chip 2 (NCFG)

Figure 27. The clock trees in the Innovus Clock Tree Debugger, opened on the CTS databases. Vertical axis: insertion delay in ns from the clock pad at the top. Yellow triangles are clock buffers, green circles the integrated clock gates, red marks the sinks. Every sink sits between 1.0 and 1.1 ns after the pad, in both chips.

4.2.11 Routing and extraction

Metals 1 to 7, 21.7 mm of wire on Chip 1 (CFG) and 12.0 mm on Chip 2 (NCFG), routing DRC clean (Fig. 28). The parasitics are then extracted per corner into SPEF files, the fast corner at −40 °C and the slow corner at 125 °C, for the power analysis and the timing sign-off that follow.

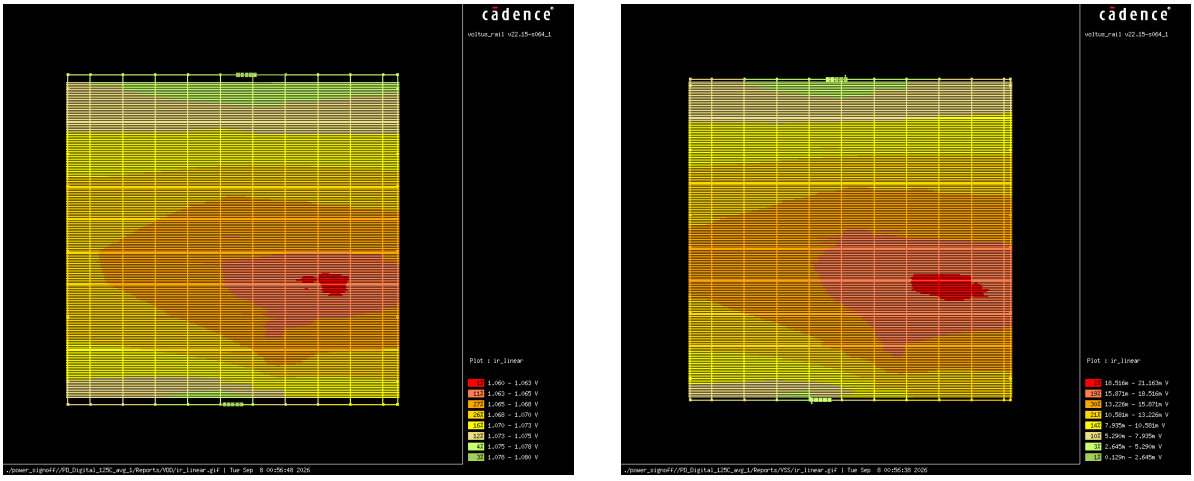


(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 28. Both chips routed and optimised, from the post-route databases.

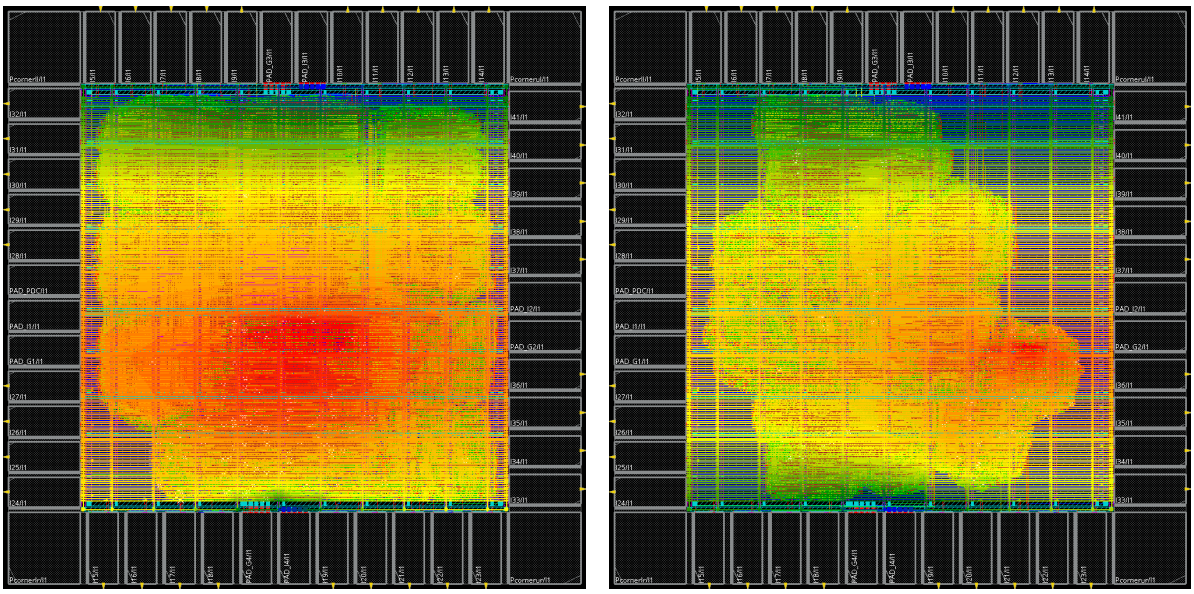
4.2.12 Power and IR drop

Voltus on the sign-off database with the switching activity of the bit-exact simulation, never vectorless: the per-block power of Chapter 5 and the rail maps of Figures 29 and 30.



(a) VDD **(b)** VSS

Figure 29. Chip 2 (NCFG) IR-drop maps from the Voltus rail analysis at the slow corner, activity from the bit-exact simulation.



(a) Chip 1 (CFG), worst 47.9 mV

(b) Chip 2 (NCFG), worst 21.2 mV

Figure 30. The VDD rail results of the same Voltus run drawn over the layout in the Innovus GUI, sign-off database of each chip. Blue is the smallest drop and red the largest: in Chip 1 (CFG) the centre of the core, in Chip 2 (NCFG) the hybrid cores on the right. The worst values are the ones in Table 5.

4.2.13 Timing sign-off

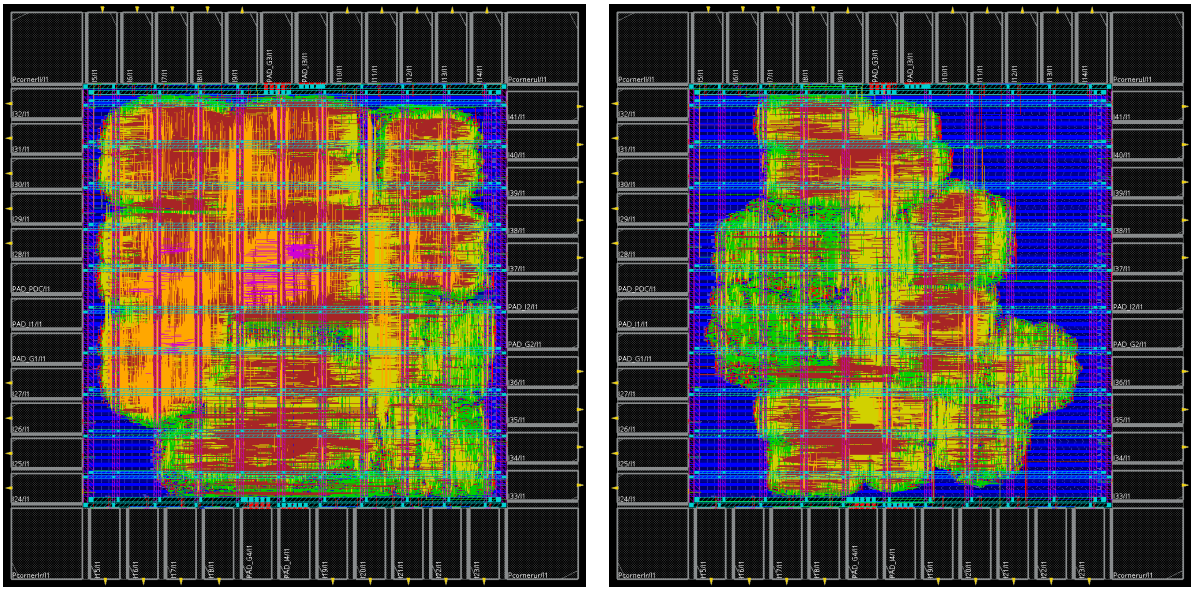
Innovus times the routed design against constraints one cycle tighter than sign-off, so that sign-off itself has margin. PrimeTime then signs off the post-layout netlist with the extracted parasitics of each corner, propagated clocks and signal integrity on, three corners, setup and hold: no path misses its setup or hold time at any corner on either chip (Table 4).

Table 4. PrimeTime sign-off on the post-layout netlist with the extracted parasitics of each corner, propagated clocks and signal integrity on: paths that miss their setup or hold time, per corner. The constraints carry a guard band of 130 ps on setup and 50 ps on hold on top of the library times. TOOL-REPORT (PrimeTime, 7 September 2026).

chip	corner	setup	hold	result
Chip 1 (CFG)	slow	0	0	pass
	typical	0	0	pass
	fast	0	0	pass
Chip 2 (NCFG)	slow	0	0	pass
	typical	0	0	pass
	fast	0	0	pass

4.2.14 Fill stage

Filler cells close every gap in the rows, the database is saved in that state, and the post-layout netlist and the SDF are written from it. The netlist carries an identity, its clock-gate and flip-flop counts, that must match the synthesised netlist before it is used (Fig. 31).



(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 31. The sign-off databases of both chips with the filler cells in place: the state that was streamed to GDS.

4.2.15 Equivalence check, gate to gate

After the fill stage Conformal compares the synthesised netlist against the post-layout netlist: PASS, zero non-equivalent points, every output and every mapped flip-flop equivalent (Figures 32 and 33).

```
================================================================================
                            Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                        0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                   0
--------------------------------------------------------------------------------
3. User modification to design:                                               0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:   0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                          1
     Black box due to undefined cells:                          yes
--------------------------------------------------------------------------------
6. Compare Results:                                                        PASS
================================================================================
```

Figure 32. The Conformal verification report of the gate-to-gate equivalence check, the synthesised netlist against the post-layout netlist: compare result PASS. The one design ambiguity is the six pad cells, which have no logic model and are black-boxed on both sides. The report is identical for the two chips.

Conformal, compared points

```
0 Non-equivalent point(s) reported
0 compared point(s) reported
================================================================================
Compared points      PO     DFF       Total
--------------------------------------------------------------------------------
Equivalent           33     2236      2269
================================================================================
```

Conformal, mapping and compare statistics

```
Mapping and compare statistics
================================================================================
                    Compare Result       Golden            Revised
--------------------------------------------------------------------------------
Root module name                         top               top

Primary inputs                              4                 4
   Mapped                                   4                 4

Primary outputs                            33                33
   Mapped                                  33                33
      Equivalent               33

Black-box key points                       13                13
   Unmapped                                13                13
      Unreachable                          13                13

State key points                         2314              2314
   Mapped                                2236              2236
      Equivalent            2236
   Unmapped                                78                78
      Unreachable                          78                78
================================================================================
```

Conformal, black boxes (pad cells without a logic model)

```
SYSTEM: (G R) PCORNER
SYSTEM: (G R) PVDD1CDG
SYSTEM: (G R) PVDD2CDG
SYSTEM: (G R) PVDD2POC
SYSTEM: (G R) PVSS1CDG
SYSTEM: (G R) PVSS2CDG
```

(a) Chip 1 (CFG): 33 outputs and 2 236 flip-flops equivalent

Conformal, compared points

```
0 Non-equivalent point(s) reported
0 compared point(s) reported
================================================================================
Compared points      PO     DFF       Total
--------------------------------------------------------------------------------
Equivalent           33     1496      1529
================================================================================
```

Conformal, mapping and compare statistics

```
Mapping and compare statistics
================================================================================
                    Compare Result       Golden            Revised
--------------------------------------------------------------------------------
Root module name                         top               top

Primary inputs                              4                 4
   Mapped                                   4                 4

Primary outputs                            33                33
   Mapped                                  33                33
      Equivalent               33

Black-box key points                       13                13
   Unmapped                                13                13
      Unreachable                          13                13

State key points                         1530              1530
   Mapped                                1496              1496
      Equivalent            1496
   Unmapped                                34                34
      Unreachable                          34                34
================================================================================
```

Conformal, black boxes (pad cells without a logic model)

```
SYSTEM: (G R) PCORNER
SYSTEM: (G R) PVDD1CDG
SYSTEM: (G R) PVDD2CDG
SYSTEM: (G R) PVDD2POC
SYSTEM: (G R) PVSS1CDG
SYSTEM: (G R) PVSS2CDG
```

(b) Chip 2 (NCFG): 33 outputs and 1 496 flip-flops equivalent

Figure 33. The Conformal results behind the PASS: every primary output and every mapped flip-flop equivalent, zero non-equivalent points. The unmapped state points are the flip-flops inside the black-boxed pad cells and the unreachable ones behind them; the six black boxes are the power, ground, corner and power-on-control pads, which have no logic model.

4.2.16 Post-layout SDF simulation, the acceptance test

The routed netlist with its SDF is simulated at both corners, all three cores, three clocks (720, 900 and 960 MHz), with timing checks off and on: 18 of 18 bit-exact, zero unknowns, both chips (Figures 34

and 35). This is the test that admits a design to the back end.

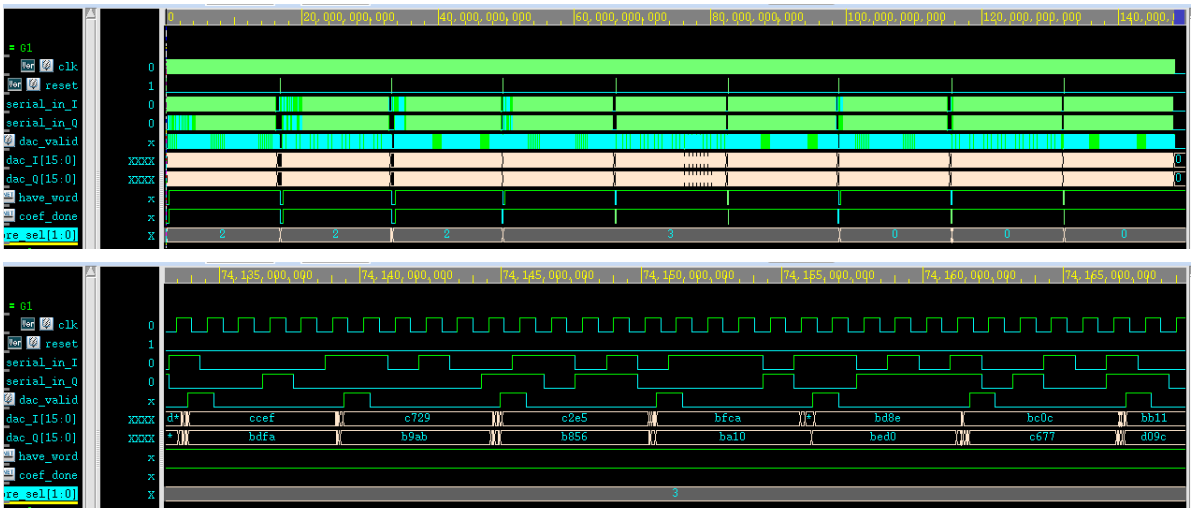


Figure 34. Chip 1 (CFG) post-layout simulation at the slow corner with timing checks on. Top: nine segments, `core_sel` stepping through the three cores. Bottom: output words inside a 960 MHz segment; the bus changes on the `dac_valid` strobe and the annotated delays are the short transition regions.

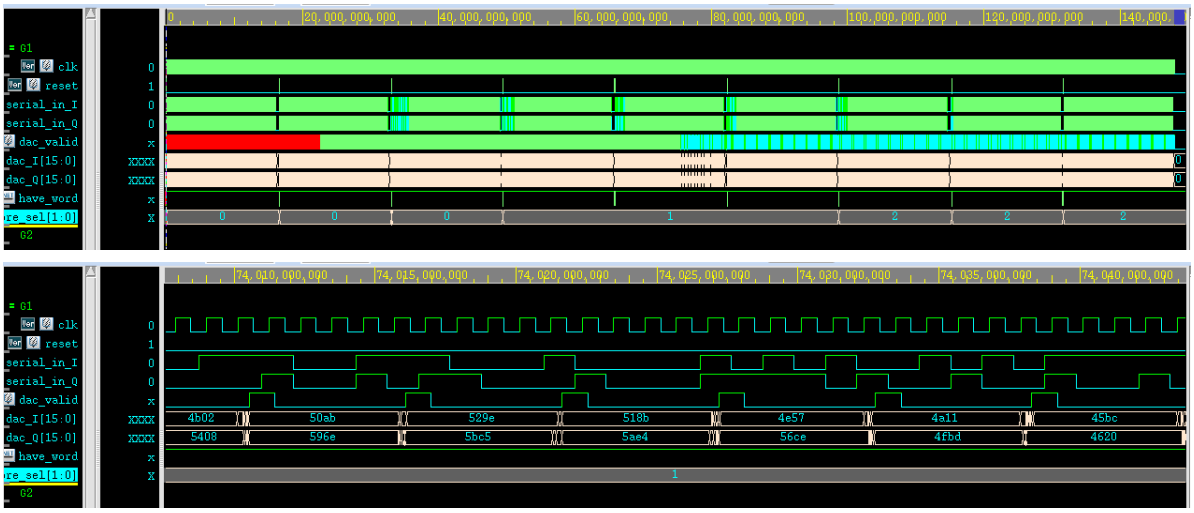
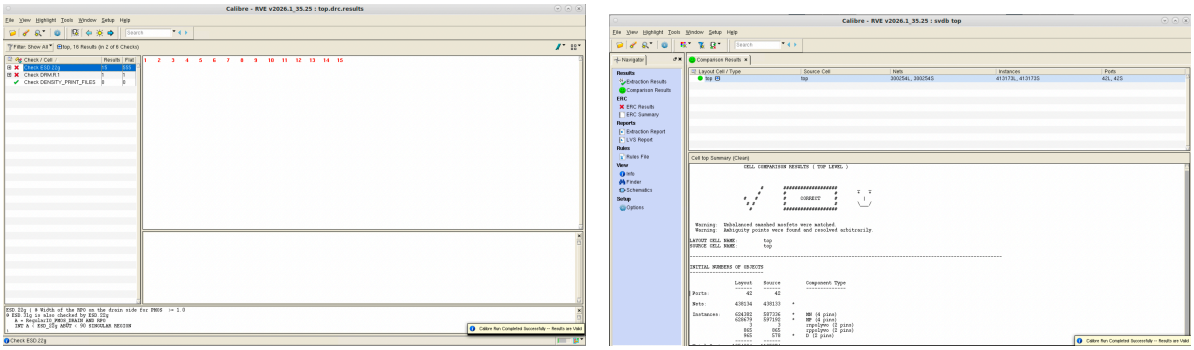


Figure 35. Chip 2 (NCFG), the same two views at the slow corner: the same segment set and the same golden words.

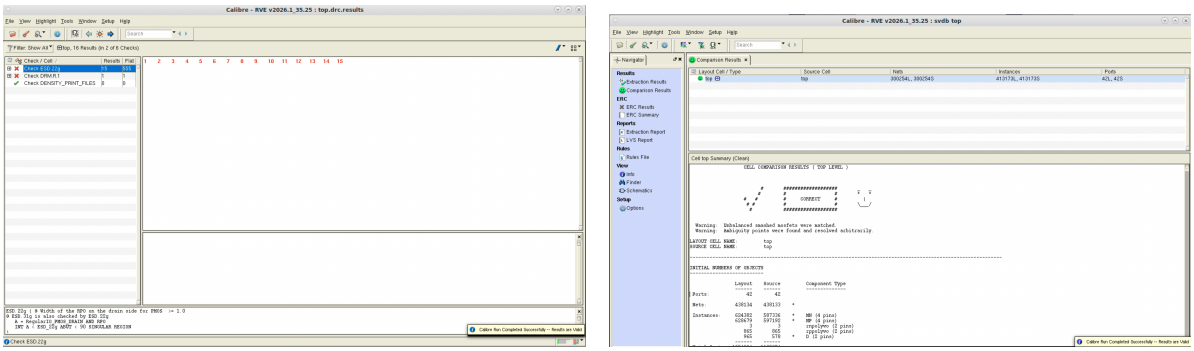
4.2.17 Physical verification in Virtuoso

The Innovus GDS is imported into Virtuoso and the power-pad pins are shorted for the netlist. Calibre then runs LVS against the full GDS and DRC: LVS CORRECT with 42 of 42 ports, ERC clean, and 16 known DRC results in two checks, waived under the rules of this pad library, on both chips (Figures 36 and 37). The 46 bond pads and the logo are added next. Then comes the dummy insertion in two passes, first the dummy metal on every routing layer, then the dummy poly and diffusion, each followed by a DRC run, so that the density rules of the process are met everywhere on the die. The filled GDS with the bond pads and the dummies is the one that goes out (Fig. 38).



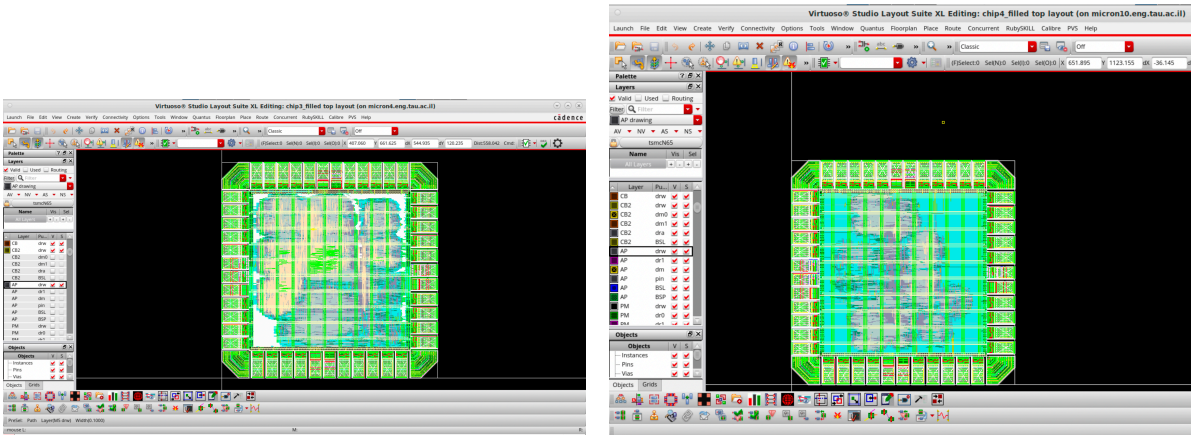
(a) Calibre RVE, filled Chip 1 (CFG) GDS **(b)** LVS on Chip 1 (CFG)

Figure 36. Physical verification on Chip 1 (CFG): 16 DRC results in 2 of 6 checks, both known and waived; LVS CORRECT, 42 ports matched.



(a) Calibre RVE, filled Chip 2 (NCFG) GDS **(b)** LVS on Chip 2 (NCFG)

Figure 37. Physical verification on Chip 2 (NCFG): the same 16 DRC results in the same 2 of 6 checks, known and waived; LVS CORRECT, 42 ports matched, 413 173 instances on both sides.



(a) Chip 1 (CFG) **(b)** Chip 2 (NCFG)

Figure 38. The filled layouts: pad ring, bond pads and dummy fill; the core is the dense region inside the ring.

4.2.18 Final GDS streamout

Both filled GDS files were submitted through the multi-project-wafer channel on 8 September 2026 and accepted for fabrication.

Chapter 5

Results

Table 5 puts the two chips side by side at sign-off: removing the multipliers and the coefficient bank takes out 29 % of the cells and 20 % of the cell area, shortens the critical path by seven levels of logic, and halves the worst IR drop; both chips are bit-exact in every post-layout run. Power (Table 6) is a Voltus report annotated with the activity of the bit-exact simulation, never vectorless; the cleanup filter dominates the loadable chip and shrinks four to five times when hardwired. Earlier analyser measurements on the same interpolator arithmetic ranked the polyphase core first on image rejection at every factor once its band edges were chosen per L; Fig. 39 shows what the analyser sees and what the cleanup filter does.

Table 5. Both chips at sign-off. Synthesis rows are TOOL-REPORT (Design Compiler); physical rows are TOOL-REPORT (Innovus, Calibre, Conformal, Voltus, PrimeTime); simulation rows are SIM-COMPARE.

	Chip 1 (CFG)	Chip 2 (NCFG)
coefficients	loadable over a serial bus	constants in the logic (SPT / CSD)
leaf cells	54 665	38 749
cell area (µm^2)	297 808	236 793
logic levels on the critical path	40	33
die / core allocation	960×960 µm, same 461 448 µm^2 core	
core utilisation	64.4 %	51.2 %
sign-off timing (PrimeTime, 3 corners)	setup and hold met at every corner	setup and hold met at every corner
DRC after fill	16 (ESD.22g 15, DRM.R.1 1; waived)	16, same split
LVS	CORRECT, 42/42 ports	CORRECT, 42/42
LEC, RTL to gate and gate to gate	PASS and PASS, 2269 points	PASS and PASS, 1529 points
worst IR drop, slow corner	47.9 mV	21.2 mV
post-layout SDF, 2 corners × 3 rates × 3 cores	18/18 bit-exact	18/18 bit-exact
signoff GDS	118 835 746 B	78 403 486 B

5.1 Power, area and EVM per core

Table 6 gives the core power of every interpolator in every mode, from the Voltus run on the routed sign-off database with the switching activity of the bit-exact simulation. The hardwired chip draws about 40 % less on the polyphase path and 50 to 60 % less on the cubic and spline paths. The die total is dominated by the 33 output pads and is nearly the same for both chips. Table 7 shows where the power goes at $L = 4$: in Chip 1 (CFG) the cleanup filter costs six times the cubic or spline core it follows, and hardwiring its taps in Chip 2 (NCFG) cuts it by four. Table 8 compares the cell area of the four arithmetic blocks: the hardwired polyphase and cleanup are 40 % and 32 % smaller, the hybrid cubic and spline 9 % and 17 % larger, and the chip as a whole 20 % smaller. Table 9 gives the EVM of every interpolator in every mode as read on the analyser: the polyphase FIR is the best core in both chips, and

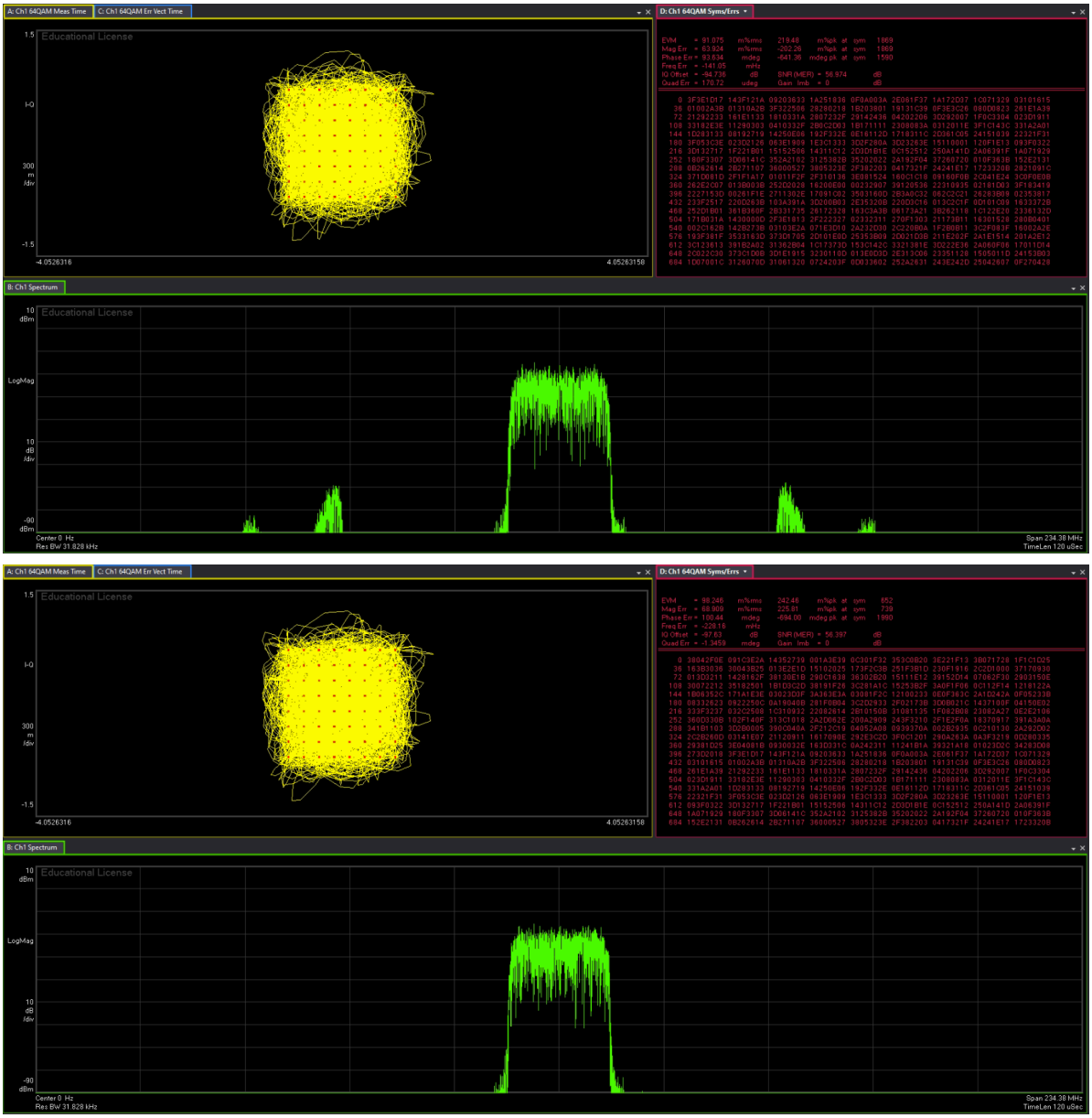


Figure 39. Keysight 89600 VSA on the RTL output stream of the spline core, an earlier iteration of the same interpolator arithmetic. Each panel: the QAM-64 constellation and error vector (top left), the demodulation readout with EVM and MER (top right), and the spectrum (bottom). Top: cleanup filter off; the two image replicas stand beside the wanted band, EVM 91 m % and MER 57 dB as read on the instrument. Bottom: cleanup filter on; the replicas are gone and only the wanted band remains, EVM 98 m % and MER 56 dB. The constellation demodulates cleanly either way: the filter's job is out of band.

hardwiring its taps does not cost EVM.

Table 6. Core power in mW, every interpolator in every mode, and the saving of Chip 2 (NCFG) over Chip 1 (CFG). TOOL-REPORT (Voltus per-block reports on the sign-off database).

	Chip 1 (CFG)			Chip 2 (NCFG)			saving		
mode	poly	cubic	spline	poly	cubic	spline	poly	cubic	spline
$L = 2$	14.52	21.02	21.27	8.80	9.81	10.46	39 %	53 %	51 %
$L = 3$	14.55	25.88	26.42	8.70	12.50	13.58	40 %	52 %	49 %
$L = 4$	16.59	30.73	30.88	9.76	13.99	13.47	41 %	54 %	56 %
$L = 5$	16.68	35.30	35.53	9.63	14.07	14.48	42 %	60 %	59 %
die with pads, $L = 4$	227.1	240.7	241.3	220.4	223.8	223.8			

Table 7. Power of every block in mW at $L = 4$. TOOL-REPORT (Voltus, the same reports).

	Chip 1 (CFG)			Chip 2 (NCFG)		
block	poly	cubic	spline	poly	cubic	spline
the selected core	8.87	2.81	3.15	4.02	2.44	2.04
cleanup filter	0.33	19.25	19.28	0.17	4.91	4.92
control unit	1.49	1.49	1.50	1.13	1.12	1.12
converter stage	0.56	0.45	0.45	0.44	0.44	0.44
the two idle cores	2.08	3.44	3.32	2.12	3.16	3.02
clock tree and glue	3.26	3.29	3.18	1.88	1.92	1.93
core total	16.59	30.73	30.88	9.76	13.99	13.47

Table 8. Cell area of the four arithmetic blocks in µm^2, from the hierarchical area report of the synthesised netlist that went to place-and-route. TOOL-REPORT (Design Compiler).

block	Chip 1 (CFG)	Chip 2 (NCFG)	change
polyphase	156 105	93 846	−40 %
cubic	45 249	49 326	+9 %
spline	55 990	65 404	+17 %
cleanup filter	36 852	25 215	−32 %
whole core	297 808	236 793	−20 %

Table 9. EVM in m % rms of every interpolator at $L = 2, 3, 4, 5$, read from the Keysight 89600 VSA on the RTL output streams. INSTRUMENT (project chip-comparison report).

chip	interpolator	$L = 2$	$L = 3$	$L = 4$	$L = 5$
Chip 1 (CFG)	polyphase	60	66	73	59
	cubic	84	127	158	129
	spline	105	142	145	125
Chip 2 (NCFG)	polyphase, hardwired	54	91	80	56
	cubic, hybrid	115	114	167	152
	spline, hybrid	148	194	146	186

Chapter 6

Ongoing work and conclusions

6.1 Ongoing work

Silicon. The dies return bare. The bench keeps every signal 3.3 V and single-ended: a chip-on-board carrier with the die wire-bonded to it, the four inputs from a pattern generator at a reduced clock for the functional tests and from a Keysight M8190A at 720, 900 and 960 MHz for the speed test, the 33 outputs captured on `dac_valid` by a state-mode logic analyser. Each die is driven from the same stimulus files at each of the seven legal modes; the acceptance criterion is unchanged, word-for-word agreement with the golden, then EVM and replica level from the instrument. **A learned interpolation kernel.** A transposed convolution with stride L is a polyphase interpolator, so the prototype can be trained; a kernel trained and quantised to five signed-power-of-two terms per tap exists beside the equiripple design, and the loadable core accepts it with no change to the silicon. **A Farrow core** is the first candidate for a successor die under the same interface and golden model.

6.2 Conclusions

The three methods inside Chip 1 (CFG). With the same loadable interface, the polyphase FIR is the best of the three on every count that matters at the output: at $L = 4$ it draws 16.6 mW against 30.7 mW for the cubic and 30.9 mW for the spline, and its EVM is the lowest at every factor (Tables 6 and 9). Its price is area: 156 105 µm^2, three times the cubic core and almost three times the spline, because it keeps 35 multipliers per rail. The cubic is the smallest and simplest core, four products per output, and the spline the smoothest interpolant in time, which pays at the two higher factors; but both need the cleanup filter at the output rate, and that filter alone costs more power than the core it follows, which is why neither can beat the polyphase on power or EVM.

The three methods inside Chip 2 (NCFG). Hardwiring keeps the ranking and moves every number: the polyphase falls to 9.8 mW and 93 846 µm^2, now 40 % smaller than its loadable self, at EVM in the same range; the hybrid cubic and spline fall to 14.0 mW and 13.5 mW, mostly because the hardwired cleanup filter costs a quarter of the loadable one, while their cores grow by 9 and 17 % for the shift paths that sit beside the multiplier they still need at $L = 3$ and $L = 5$. The polyphase remains the method to choose on this chip too, and here it is also the only core that needs no multiplier at all.

Chip 1 (CFG) against Chip 2 (NCFG). At the same modes and the same EVM, Chip 2 (NCFG) draws 39 to 60 % less core power and occupies 20 % less area (Tables 6 and 8). What Chip 1 (CFG) buys with its extra silicon is freedom: every tap is loaded over the serial bus, so the passband and the band edges of the interpolation filter, the cleanup taps, and therefore the symbol rate the chip serves, can all be changed in the field or retuned on the bench. Chip 2 (NCFG) has its taps in the logic: its passband and its symbol rate are fixed at the ones it was designed for. Both chips keep the same operating freedom otherwise, since the clock frequency and the interpolation factor come from the opcode, not from the taps: any of the seven modes, three clocks and four factors, runs on either chip. The verdict is therefore a question of use. For a product with one known symbol rate, the hardwired polyphase of Chip 2

(NCFG) is the best design in this work, the lowest power and the smallest area at the best EVM. For a laboratory chip, a multi-standard transmitter, or any case where the band may still move, the loadable polyphase of Chip 1 (CFG) is the right one, and the cubic and spline cores remain the fallback when area is the only constraint and a cleanup filter can be afforded.

The best design. Taking the two chips and the six cores together, the best design in this work is the hardwired polyphase FIR of Chip 2 (NCFG). It has the lowest power of any core at every factor, 8.8 to 9.8 mW against 14 to 36 mW for every other combination; it is 40 % smaller than the loadable polyphase and needs no multiplier and no coefficient bus; its EVM is the lowest on its chip at every factor and in the same range as the loadable polyphase; and it is bit-exact to the golden model at every mode and both corners. Its one limitation, a fixed passband and therefore a fixed symbol rate, is the price of all of that, and for a transmitter built to one standard it is no limitation at all. Second comes the loadable polyphase of Chip 1 (CFG), the same arithmetic at twice the power and area in exchange for a band that can be moved. The cubic and the spline, on either chip, are the right choice only where the core must be as small as possible and the power of the cleanup filter can be accepted.

1. The polyphase FIR needs no cleanup filter, and choosing its band edges per interpolation factor bought 7.7 dB of image rejection at $L = 4$ with EVM improved at the same time, at zero hardware cost.
2. Hardwiring the coefficients removes every multiplier from the polyphase and cleanup datapaths and gives a chip 29 % smaller in cells and 20 % in area while remaining bit-exact to the same golden model.
3. Both chips are bit-exact in post-layout simulation at 720, 900 and 960 MHz at both corners, pass DRC, LVS and formal equivalence, and are in fabrication.
4. The method, one golden model, a bit-exact compare at every stage, and no number accepted from a table that a simulation has not confirmed, is the most transferable result of the work.

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תקציר

משדר ספרתי המעביר דגימות QAM-64 לממיר בקצב של 60 מיליון דגימות בשנייה משכפל את הספקטרום שלו כל 60 מגה-הרץ, והמסנן האנלוגי שצריך להסיר את התמונה הראשונה הוא תלול, יקר ורגיש לטמפרטורה. אינטרפולציה ספרתית לפני הממיר מרחיקה את התמונה הזו ומחליפה את הבעיה האנלוגית באריתמטיקה ספרתית --- ובפרט במכפלים של מסנן האינטרפולציה. עבודה זו עוסקת באריתמטיקה הזו: איזו ארכיטקטורת אינטרפולציה נותנת את ה-EVM ואת רמת הרפליקה הטובים ביותר עבור מספר מכפלים נתון, מה עולה להסיר את המכפלים לחלוטין על-ידי קיבוע המקדמים כעצי הזזה-וחיבור, וכיצד מובל תכן ממודל מתמטי לסיליקון חתום בלי מספר אחד ברשומה שאינו ניתן למעקב עד לדוח הכלי או למדידה שממנה הגיע.

שלוש ארכיטקטורות אינטרפולציה --- מסנן FIR פוליפאזי, אינטרפולטור קובי, וספליין הרמיט רקורסיבי מסוג MinAJ2 --- חולקות כל אחת משתי פיסות סיליקון בתהליך TSMC 65 ננומטר ,LP מאחורי ממשק אחד הניתן לקינפוג בזמן ריצה, המכסה ארבעה מקדמי אינטרפולציה ושלושה אורכי מילה בתדר של עד 960 מגה-הרץ. הפיתוח המתמטי כולל את הפירוק הפוליפאזי, תכן קצוות הפס לכל מקדם אינטרפולציה של הפרוטוטיפ בעל האדווה השווה, הגרעין הקובי וגרעין הרמיט עם טבלאות המשקלים בנקודה קבועה, חסמי הצוברים שנגזרו לכל ליבה, והחלפת המכפלים בסכומי חזקות שתיים מסומנות עם קריטריון הדיוק הדיאדי שלה; כל משוואה נבדקה מול מודל ה-MATLAB ומול ה-RTL שמהם נבנו השבבים. שבב אחד שומר על כל מקדם כטעין; השני מקבע כל מקדם, והוא קטן יותר בכל מדד סינתזה.

זרימת התכן מתועדת שלב אחר שלב --- מודל הזהב ב-MATLAB, ,RTL סינתזה, טבעת הפדים, תכנון הרצפה, מיקום, עץ השעון, ניתוב, תזמון לאחר פריסה, סימולציית SDF לאחר פריסה, ,LVS ,DRC בדיקת שקילות, נפילת מתח, חתימת תזמון והזרמת GDS --- כאשר הדוח של כל שלב משוחזר מפלט הכלי עצמו והפריסה נלכדת ממסדי הנתונים הסופיים. מודל זהב אחד מושווה ביט אחר ביט בכל ייצוג של התכן, ושני השבבים מדויקים-ביט בסימולציה לאחר פריסה בכל מצב עבודה חוקי ובשתי פינות האפיון; שניהם הוגשו לייצור בספטמבר 2026.

כל מספר במסמך זה ניתן למעקב בעבודת הגמר עד לדוח, ליומן, להשוואת הסימולציה או לקריאת המכשיר שממנו נקרא; אף מספר אינו הערכה, ולא נטענת שום תוצאה מסיליקון מדוד.

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אריתמטיקה בהספק נמוך למסנני אינטרפולציה דיגיטליים רב-קצביים הניתנים להגדרה, מ-MATLAB ועד GDSII: השוואת פוליפאזי, קובי וספליין על שני טייפ-אאוטים בתהליך TSMC 65 ננומטר

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